SOURCE CODE REPORT

Data Analytics Implementation

System Deployment:

https://nhom10trdevidedby5.shinyapps.io/thi_data/

NOTE: Shinyapps free tier. Please wait 30-60 seconds for server to start.

Source: app.R

```
library(shiny)
library(shinydashboard)
library(dplyr)
library(tidyr)
library(lubridate)
library(ggplot2)
library(plotly)
library(DT)
library(stats)
library(readxl)
library(scales)

for (i in 1:10) {
  file_name <- paste0("phase", i, ".R")
  if (file.exists(file_name)) {
    source(file_name, local = TRUE)
  } else {
    warning(paste("Khong tim thay file:", file_name))
  }
}

GLOBAL_DATA_CACHE <- NULL
if (file.exists("data_final.rds")) {
  GLOBAL_DATA_CACHE <- readRDS("data_final.rds")
}

COLORS <- list(
  primary   = "#1F3864",
  secondary = "#4472C4",
  accent    = "#1F3864",

  palette = c("#4472C4", "#9467BD", "#2CA02C", "#D62728", "#5B9BD5", "#FF7F0E", "#8C564B"),

  heat_low  = "#D9D7F1",
  heat_high = "#1F3864",

  success = "#2CA02C",
  danger  = "#D62728",
  warning = "#FF7F0E",
  info    = "#5B9BD5"
)

format_vnd <- function(x) {
  ifelse(is.na(x), "N/A",
    ifelse(abs(x) >= 1e9,
        paste0(formatC(x / 1e9, format = "f", digits = 1, big.mark = ".", decimal.mark = ","), "
ty"),
```

```
    ifelse(abs(x) >= 1e6,
        paste0(formatC(x / 1e6, format = "f", digits = 1, big.mark = ".", decimal.mark = ","), "
tr"),
      formatC(x, format = "f", digits = 0, big.mark = ".", decimal.mark = ",")
    ))
  )
}

label_trieu <- function(x) {
  ifelse(abs(x) >= 1e9, paste0(round(x/1e9, 1), " ty"),
    ifelse(abs(x) >= 1e6, paste0(round(x/1e6, 1), " tr"),
      formatC(x, format = "f", digits = 0, big.mark = ".", decimal.mark = ",")))
}

theme_academic <- function(base_size = 14) {
  theme_minimal(base_size = base_size, base_family = "sans") +
    theme(
      text = element_text(family = "sans", color = "#1F3864"),
      plot.title = element_text(face = "bold", size = base_size + 2, color = COLORS$primary, hjust
= 0),
      plot.subtitle = element_text(size = base_size - 1, color = "#4472C4"),
      axis.title = element_text(face = "bold", size = base_size - 2, color = "#1F3864"),
      axis.text = element_text(size = base_size - 3, color = "#4472C4"),
      legend.title = element_text(face = "bold", size = base_size - 2, color = COLORS$primary),
      legend.text = element_text(size = base_size - 3),
      panel.grid.major.y = element_line(color = "#D9D7F1", linewidth = 0.5),
      panel.grid.major.x = element_line(color = "#D9D7F1", linewidth = 0.3),
      panel.grid.minor = element_blank(),
      panel.background = element_rect(fill = "#FFFFFF", color = NA),
      plot.background = element_rect(fill = "#FFFFFF", color = NA),
      strip.text = element_text(face = "bold", color = COLORS$primary),
      plot.margin = margin(0, 0, 0, 0)
    )
}

load_master_data <- function(excel_path) {
  message(">>> Dang doc file Excel: ", excel_path)

  don_hang    <- read_excel(excel_path, sheet = "Don hang")
  hop_dong    <- read_excel(excel_path, sheet = "Hop dong")
  khach_hang  <- read_excel(excel_path, sheet = "Khach hang")
  goi_sp      <- read_excel(excel_path, sheet = "Goi San Pham")
  nhom_sp     <- read_excel(excel_path, sheet = "Nhom san pham")
  kenh_ban    <- read_excel(excel_path, sheet = "Kenh Ban")
  kenh_ct     <- read_excel(excel_path, sheet = "Kenh ban chi tiet")
  chi_nhanh   <- read_excel(excel_path, sheet = "Chi nhanh")
  ma_nv       <- read_excel(excel_path, sheet = "Ma nhan vien")
  nhan_vien   <- read_excel(excel_path, sheet = "Nhan vien")
  do_tuoi     <- read_excel(excel_path, sheet = "Do tuoi")
  xe_cg       <- read_excel(excel_path, sheet = "Xe co gioi")
```

```
  hang_xe     <- read_excel(excel_path, sheet = "Hang Xe")
  tt_xe       <- read_excel(excel_path, sheet = "Thong tin xe co gioi")
  kpi_raw     <- read_excel(excel_path, sheet = "KPI")
  phong_ban   <- read_excel(excel_path, sheet = "Phong ban")

  master <- don_hang %>%
    left_join(goi_sp, by = c("MA_GSP" = "MA_GOISANPHAM")) %>%
    left_join(nhom_sp, by = c("MA_NHOMSANPHAM" = "MA_NHOMSANPHAM")) %>%
    left_join(kenh_ct, by = "MA_KENHBANCHITIET") %>%
    left_join(kenh_ban, by = "MA_KENHBAN") %>%
    left_join(ma_nv %>% select(MA_NV, MA_CN, MA_PB, MA_CONGTY), by = "MA_NV") %>%
    left_join(chi_nhanh %>% select(MA_CN, TEN_CN), by = "MA_CN")

  kh_full <- khach_hang %>% left_join(do_tuoi, by = "TUOI")
  master <- master %>%
      left_join(kh_full %>% select(MA_KH, TUOI, GIOI_TINH, NGHENGHIEP, XHKH, NHOM_TUOI), by =
"MA_KH")

  xe_info <- tt_xe %>%
      left_join(xe_cg %>% select(MA_DONG_XE, TEN_DONG_XE, GIA_TIEN_XE = GIA_TIEN, MA_HANG_XE,
SO_CHO_NGOI), by = "MA_DONG_XE") %>%
    left_join(hang_xe, by = "MA_HANG_XE")
  master <- master %>%
      left_join(xe_info %>% select(MA_HD, TEN_DONG_XE, GIA_TIEN_XE, TEN_HANG_XE, BIEN_SO_XE,
SO_CHO_NGOI), by = "MA_HD")

  master <- master %>%
    mutate(
      NGAY_KY_HD    = as.Date(NGAY_KY_HD),
      NGAY_HIEU_LUC = as.Date(NGAY_HIEU_LUC),
      NGAY_HET_HAN  = as.Date(NGAY_HET_HAN),
      PHIBH          = as.numeric(PHIBH),
      GIA_BH         = as.numeric(GIA_BH),
      GIA_TIEN       = as.numeric(GIA_TIEN),
      TUOI           = as.numeric(TUOI),
      THANG          = floor_date(NGAY_KY_HD, "month"),
      THANGNAM       = format(NGAY_KY_HD, "%Y%m")
    ) %>%
    mutate(across(where(is.character), ~ ifelse(is.na(.), "Chua xac dinh", .)))

  nv_info <- ma_nv %>% left_join(nhan_vien %>% select(MA_GOCNV, HO_TEN, SDT), by = "MA_GOCNV")
  master <- master %>% left_join(nv_info %>% select(MA_NV, HO_TEN_NV = HO_TEN, SDT_NV = SDT), by =
"MA_NV")

   return(list(master = master, kpi = kpi_raw, hop_dong = hop_dong, phong_ban = phong_ban,
chi_nhanh = chi_nhanh))
}

ui <- dashboardPage(
  skin = "blue",
```

```
dashboardHeader(title = "Bao Hiem Quant Dashboard", titleWidth = 320),
dashboardSidebar(width = 320, sidebarMenu(id = "tabs",
  menuItem("Phase 1: Chat luong du lieu",     tabName = "tab_phase1", icon = NULL),
  menuItem("Phase 2: Khai pha vi mo",         tabName = "tab_phase2", icon = NULL),
  menuItem("Phase 3: Phan tich chuyen sau",   tabName = "tab_phase3", icon = NULL),
  menuItem("Phase 4: Phan tich phan khuc",    tabName = "tab_phase4", icon = NULL),
  menuItem("Phase 5: Mang luoi va van Hanh",   tabName = "tab_phase5", icon = NULL),
  menuItem("Phase 6: Du bao & rui ro",        tabName = "tab_phase6", icon = NULL),
  menuItem("Phase 7: Chien luoc",  tabName = "tab_phase7", icon = NULL),
  menuItem("Phase 8: Quan tri KPI",           tabName = "tab_phase8", icon = NULL),
  menuItem("Phase 9: KH tiem nang va Tai tuc", tabName = "tab_phase9", icon = NULL),
  menuItem("Phase 10: Tong ket",        tabName = "tab_phase10", icon = NULL)  )),

dashboardBody(
  tags$head(
    tags$script(HTML("
      $(document).on('shiny:visualchange', function(event) {
        setTimeout(function() {
          window.dispatchEvent(new Event('resize'));
          $('.js-plotly-plot').each(function() {
            Plotly.Plots.resize(this);
          });
        }, 350);
      });
    ")),

    tags$style(HTML(sprintf('
      body, .main-header .logo, .sidebar-menu, h1, h2, h3, h4, h5, h6 {
        font-family: "Helvetica Neue", Helvetica, Arial, sans-serif !important;
      }

      .plotly.html-widget, .js-plotly-plot {
        width: 100%% !important;
        height: 100%% !important;
        min-height: 350px;
      }

      .content { padding: 15px !important; }
      .box-body { padding: 10px !important; }

      .skin-blue .main-header .navbar { background-color: %s !important; }
      .skin-blue .main-header .logo {
        background-color: #1F3864 !important;
        color: #FFFFFF !important;
        font-weight: bold;
      }
      .skin-blue .main-header .logo:hover { background-color: %s !important; }

      .skin-blue .main-sidebar {
        background-color: #FFFFFF !important;
```

```
          border-right: 1px solid #D9D7F1;
        }
        .skin-blue .sidebar-menu > li > a { color: #1F3864 !important; font-size: 15px; }
        .skin-blue .sidebar-menu > li:hover > a,
        .skin-blue .sidebar-menu > li.active > a {
          background-color: #D9D7F1 !important;
          color: %s !important;
          border-left-color: %s !important;
          font-weight: bold;
        }

        .box.box-primary { border-top-color: %s !important; }
        .box.box-success { border-top-color: %s !important; }
        .box.box-danger  { border-top-color: %s !important; }
        .box.box-info    { border-top-color: %s !important; }
        .box.box-warning { border-top-color: %s !important; }

        table.dataTable thead th {
          background-color: %s !important;
          color: #FFFFFF !important;
          border-bottom: none !important;
        }
      ',
      COLORS$primary, COLORS$primary, COLORS$primary, COLORS$primary,
      COLORS$primary, COLORS$success, COLORS$danger, COLORS$info, COLORS$warning,
      COLORS$primary)))
    ),

    tabItems(
      phase1UI("tab_phase1"), phase2UI("tab_phase2"), phase3UI("tab_phase3"),
      phase4UI("tab_phase4"), phase5UI("tab_phase5"), phase6UI("tab_phase6"),
      phase7UI("tab_phase7"), phase8UI("tab_phase8"), phase9UI("tab_phase9"),
      phase10UI("tab_phase10")
    )
  )
)

server <- function(input, output, session) {
  all_data <- reactiveVal(GLOBAL_DATA_CACHE)

  if (is.null(GLOBAL_DATA_CACHE)) {
    showNotification("Loi: Du lieu chua nap vao RAM.", type = "error")
  }

  phase1Server("tab_phase1", all_data)
  phase2Server("tab_phase2", all_data)
  phase3Server("tab_phase3", all_data)
  phase4Server("tab_phase4", all_data)
  phase5Server("tab_phase5", all_data)
  phase6Server("tab_phase6", all_data)
```

```
  phase7Server("tab_phase7", all_data)
  phase8Server("tab_phase8", all_data)
  phase9Server("tab_phase9", all_data)
  phase10Server("tab_phase10", all_data)
}

shinyApp(ui = ui, server = server)
```

Source: phase1.R

```
PAL <- list(
  navy      = "#1F3864",
  steel     = "#4472C4",
  teal      = "#5B9BD5",
  purple    = "#9467BD",
  pink      = "#D9D7F1",
  skyblue   = "#5B9BD5",
  green     = "#2CA02C",
  gold      = "#FF7F0E",
  red_soft  = "#D62728",
  green_bar = "#2CA02C",
  txt_main  = "#1F3864",
  txt_light = "#4472C4",
  grid_col  = "#D9D7F1",
  bg        = "#D9D7F1"
)

theme_pptx <- function(base_size = 11.5) {
  theme_minimal(base_size = base_size, base_family = "sans") %+replace%
    theme(
      text             = element_text(family = "sans", color = PAL$txt_main),
      plot.title       = element_text(face = "bold", color = PAL$navy, size = base_size + 2, hjust
= 0,
                                      margin = margin(b = 6)),
        plot.subtitle     = element_text(color = PAL$txt_light, size = base_size - 1, margin =
margin(b = 8)),
          axis.title          = element_text(face = "bold", size = base_size - 0.5, color =
PAL$txt_main),
      axis.text        = element_text(size = base_size - 1.5, color = PAL$txt_main),
        panel.grid.major.y = element_line(color = PAL$grid_col, linetype = "dashed", linewidth =
0.35),
      panel.grid.major.x = element_blank(),
      panel.grid.minor   = element_blank(),
      legend.position  = "top",
      legend.title     = element_text(face = "bold", size = base_size - 1),
      legend.text      = element_text(size = base_size - 1.5),
      strip.text       = element_text(face = "bold"),
      plot.margin      = margin(2, 2, 2, 2)
    )
}

ly_layout <- function(p, bl = 70, bb = 40, bt = 10, br = 15) {
  p %>% layout(
    margin = list(l = bl, r = br, t = bt, b = bb),
    xaxis  = list(automargin = TRUE),
    yaxis  = list(automargin = TRUE),
    font   = list(family = "Helvetica Neue, Helvetica, Arial, sans-serif", color = PAL$txt_main)
```

```
  )
}

phase1UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,

    tags$style(HTML("
      .bg-blue { background-color: #1F3864 !important; }
      .bg-teal { background-color: #5B9BD5 !important; }
      .bg-yellow { background-color: #FF7F0E !important; }
      .bg-green { background-color: #2CA02C !important; }
      .small-box .icon-large { color: rgba(255, 255, 255, 0.2) !important; }
      .small-box { color: #FFFFFF !important; }
    ")),

    h2("Phan 1: Kiem soat chat luong va lam sach du lieu",
             style = "color:#1F3864; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
     p("Phan tich toan dien ve chat luong du lieu: Missing, Outliers, bat thuong logic va du lieu trung lap.",
                      style = "color:#4472C4; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),

    fluidRow(
      valueBoxOutput(ns("box_total_goi"), width = 3),
      valueBoxOutput(ns("box_total_hd"),  width = 3),
      valueBoxOutput(ns("box_total_kh"),  width = 3),
      valueBoxOutput(ns("box_missing_pct"), width = 3)
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Ty le thieu du lieu cac cot",
                                 style = "color:#1F3864; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Cac cot co ty le thieu cao nhat",
                                  style = "color:#4472C4; font-size:12px; margin-bottom:6px; font-family:'Helvetica Neue',sans-serif;"),
          plotlyOutput(ns("plot_missing"), height = "370px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Phan phoi phi bao hiem",
                                 style = "color:#1F3864; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Ngoai lai theo quy tac IQR x 1.5.",
                                  style = "color:#4472C4; font-size:12px; margin-bottom:6px; font-family:'Helvetica Neue',sans-serif;"),
          plotlyOutput(ns("plot_outlier"), height = "370px"))
    ),
```

```
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Thong ke phi bao hiem theo nhom san pham",
                                   style = "color:#1F3864; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("So sanh phan phoi phi.",
                                    style = "color:#4472C4; font-size:12px; margin-bottom:6px;
font-family:'Helvetica Neue',sans-serif;"),
          plotlyOutput(ns("plot_outlier_by_group"), height = "370px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Phan phoi gia tri bao hiem",
                                   style = "color:#1F3864; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("",
                                    style = "color:#4472C4; font-size:12px; margin-bottom:6px;
font-family:'Helvetica Neue',sans-serif;"),
          plotlyOutput(ns("plot_gia_bh_dist"), height = "370px"))
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Mat do phi bao hiem moi va tai tuc",
                                   style = "color:#1F3864; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("So sanh phan phoi giua hop dong moi va tai tuc",
                                    style = "color:#4472C4; font-size:12px; margin-bottom:6px;
font-family:'Helvetica Neue',sans-serif;"),
          plotlyOutput(ns("plot_density_overlay"), height = "370px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Bien dong SL goi BH theo thang",
                                   style = "color:#1F3864; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("",
                                    style = "color:#4472C4; font-size:12px; margin-bottom:6px;
font-family:'Helvetica Neue',sans-serif;"),
          plotlyOutput(ns("plot_mom_change"), height = "370px"))
    ),

    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Chat luong du lieu dau vao",
                                   style = "color:#1F3864; font-weight:700; margin:0 0 6px 0;
font-family:'Helvetica Neue',sans-serif;"),
          DTOutput(ns("table_quality_report")))
    ),

    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Nhat ky Audit: Hop dong bat thuong va ngoai lai",
                                   style = "color:#1F3864; font-weight:700; margin:0 0 6px 0;
```

```
font-family:'Helvetica Neue',sans-serif;"),
          DTOutput(ns("table_audit")))
    )
  )
}

phase1Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {

    get_data <- reactive({ req(all_data()); all_data()$master })

    output$box_total_goi <- renderValueBox({
      valueBox(format(nrow(get_data()), big.mark = "."), "Tong so goi BH",
               icon = icon("database"), color = "blue")
    })
    output$box_total_hd <- renderValueBox({
      valueBox(format(n_distinct(get_data()$MA_HD), big.mark = "."), "So luong hop dong",
               icon = icon("file-contract"), color = "blue")
    })
    output$box_total_kh <- renderValueBox({
      valueBox(format(n_distinct(get_data()$MA_KH), big.mark = "."), "Khach hang unique",
               icon = icon("users"), color = "teal")
    })
    output$box_missing_pct <- renderValueBox({
      df <- get_data()
      total_cells <- nrow(df) * ncol(df)
      na_n <- sum(is.na(df))
       cxd_n <- sum(sapply(df, function(x) if(is.character(x)) sum(x == "Chua xac dinh", na.rm =
TRUE) else 0))
      pct <- round((na_n + cxd_n) / total_cells * 100, 2)
      valueBox(paste0(pct, "%"), "Ty le missing",
               icon = icon("exclamation-triangle"), color = ifelse(pct > 5, "yellow", "green"))
    })

    output$plot_missing <- renderPlotly({
      df <- get_data()
      miss_df <- data.frame(
        Cot = names(df),
        NA_Count = colSums(is.na(df)),
         CXD_Count = sapply(df, function(x) if(is.character(x)) sum(x == "Chua xac dinh", na.rm =
TRUE) else 0),
        stringsAsFactors = FALSE
      ) %>%
        mutate(Total = NA_Count + CXD_Count, Pct = round(Total / nrow(df) * 100, 2)) %>%
        filter(Total > 0) %>% arrange(desc(Total)) %>% head(15)

      if (nrow(miss_df) == 0) miss_df <- data.frame(Cot = "Sach 100%", Total = 0, Pct = 0)

      p <- ggplot(miss_df, aes(x = reorder(Cot, Total), y = Pct)) +
        geom_col(fill = PAL$steel, width = 0.72) +
```

```
        geom_text(aes(label = paste0(Pct, "%")), hjust = -0.08, size = 3.3,
                  color = PAL$navy, fontface = "bold") +
        coord_flip() +
        scale_y_continuous(expand = expansion(mult = c(0, 0.18))) +
        theme_pptx() +
        theme(panel.grid.major.y = element_blank(),
                    panel.grid.major.x = element_line(color = PAL$grid_col, linetype = "dashed",
linewidth = 0.3)) +
        labs(x = NULL, y = "Ty le Missing (%)")

      ggplotly(p, tooltip = c("y")) %>% ly_layout(bl = 100)
    })

    output$plot_outlier <- renderPlotly({
      df <- get_data() %>% filter(PHIBH > 0)

      n_out <- {
        q1 <- quantile(df$PHIBH, 0.25); q3 <- quantile(df$PHIBH, 0.75)
        iqr <- q3 - q1; sum(df$PHIBH > q3 + 1.5 * iqr | df$PHIBH < q1 - 1.5 * iqr)
      }

      p <- ggplot(df, aes(y = PHIBH, x = "Phi BH")) +
        geom_boxplot(fill = PAL$skyblue, color = PAL$navy, width = 0.45,
                     outlier.colour = PAL$pink, outlier.size = 1.3) +
        scale_y_log10(labels = label_trieu) +
        theme_pptx() +
        theme(axis.text.x = element_text(face = "bold", size = 11)) +
        labs(x = NULL, y = "Phi BH",
             subtitle = paste0(format(n_out, big.mark = "."), " outliers detected (IQR x 1.5)"))

      ggplotly(p) %>% ly_layout(bl = 70, bb = 35)
    })

    output$plot_outlier_by_group <- renderPlotly({
      df <- get_data() %>% filter(PHIBH > 0, !is.na(NHOMSANPHAM))

        pal6 <- c(PAL$navy, PAL$steel, PAL$teal, PAL$purple, PAL$green, PAL$gold, PAL$red_soft,
PAL$pink)

      p <- ggplot(df, aes(x = NHOMSANPHAM, y = PHIBH, fill = NHOMSANPHAM)) +
        geom_boxplot(color = PAL$navy, linewidth = 0.35,
                     outlier.colour = PAL$pink, outlier.size = 0.7) +
        scale_fill_manual(values = pal6) +
        scale_y_log10(labels = label_trieu) +
        theme_pptx() +
        theme(axis.text.x = element_text(angle = 22, hjust = 1, face = "bold"),
              legend.position = "none") +
        labs(x = NULL, y = "Phi BH (VND)  Log")

      ggplotly(p) %>% ly_layout(bl = 65, bb = 70)
```

```
})

output$plot_gia_bh_dist <- renderPlotly({
  df <- get_data() %>% filter(GIA_BH > 0)

  p <- ggplot(df, aes(x = GIA_BH)) +
    geom_histogram(bins = 45, fill = PAL$teal, color = "#FFFFFF", linewidth = 0.2) +
    scale_x_log10(labels = label_trieu) +
    scale_y_continuous(expand = expansion(mult = c(0, 0.08)),
                       labels = function(x) format(x, big.mark = ".")) +
    theme_pptx() +
    theme(panel.grid.major.x = element_blank()) +
    labs(x = "Gia tri BH (VND)  Log", y = "So luong")

  ggplotly(p) %>% ly_layout(bl = 55, bb = 42)
})

output$plot_density_overlay <- renderPlotly({
  df <- get_data() %>% filter(PHIBH > 0, LOAI_HD %in% c("Moi", "Tai tuc"))

  p <- ggplot(df, aes(x = PHIBH, fill = LOAI_HD, color = LOAI_HD)) +
    geom_density(linewidth = 1) +
    scale_x_log10(labels = label_trieu) +
    scale_fill_manual(values = c("Moi" = PAL$steel, "Tai tuc" = PAL$gold)) +
    scale_color_manual(values = c("Moi" = PAL$steel, "Tai tuc" = PAL$gold)) +
    theme_pptx() +
    theme(legend.title = element_blank(),
          panel.grid.major.x = element_blank()) +
    labs(x = "Phi bao hiem", y = "Mat do")

  ggplotly(p) %>% ly_layout(bl = 50, bb = 42)
})

output$plot_mom_change <- renderPlotly({
  df_m <- get_data() %>%
    group_by(THANG) %>% summarise(N = n(), .groups = "drop") %>%
    arrange(THANG) %>%
    mutate(Pct = round((N / lag(N) - 1) * 100, 1),
           Direction = ifelse(Pct >= 0, "Tang", "Giam"))

  df_m <- df_m %>% filter(!is.na(Pct))

  p <- ggplot(df_m, aes(x = THANG, y = Pct, fill = Direction)) +
    geom_col(width = 22) +
    geom_hline(yintercept = 0, color = PAL$navy, linewidth = 0.4) +
    scale_fill_manual(values = c("Tang" = PAL$green_bar, "Giam" = PAL$red_soft)) +
    scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +
    theme_pptx() +
    theme(legend.position = "none",
          axis.text.x = element_text(angle = 45, hjust = 1, size = 9)) +
```

```
        labs(x = NULL, y = "% Thay doi so voi thang truoc")

      ggplotly(p) %>% ly_layout(bl = 50, bb = 60)
    })

    output$table_quality_report <- renderDT({
      df <- get_data()
      report <- data.frame(
        Chi_so = c("Tong so goi bao hiem", "Tong hop dong",
                   "Tong so KH unique", "Thoi gian",
                   "Phi BH binh quan ", "Phi BH trung vi",
                   "Phi BH min", "Phi BH max",
                   "So goi phi BH am", "So goi thieu gia BH",
                   "Ty le HD moi", "Ty le tai tuc"),
        Gia_tri = c(
          paste0(formatC(nrow(df), format = "d", big.mark = "."), " goi"),
          paste0(formatC(n_distinct(df$MA_HD), format = "d", big.mark = "."), " hop dong"),
          paste0(formatC(n_distinct(df$MA_KH), format = "d", big.mark = "."), " khach hang"),
           paste(format(min(df$NGAY_KY_HD, na.rm = T), "%d/%m/%Y"), "->", format(max(df$NGAY_KY_HD,
na.rm = T), "%d/%m/%Y")),
          format_vnd(mean(df$PHIBH, na.rm = T)),
          format_vnd(median(df$PHIBH, na.rm = T)),
          format_vnd(min(df$PHIBH, na.rm = T)),
          format_vnd(max(df$PHIBH, na.rm = T)),
          formatC(sum(df$PHIBH <= 0, na.rm = T), format = "d", big.mark = "."),
          formatC(sum(is.na(df$GIA_BH)), format = "d", big.mark = "."),
          paste0(round(sum(df$LOAI_HD == "Moi", na.rm = T) / nrow(df) * 100, 1), "%"),
          paste0(round(sum(df$LOAI_HD == "Tai tuc", na.rm = T) / nrow(df) * 100, 1), "%")
        ), stringsAsFactors = FALSE
      )

      datatable(report,
        colnames = c("Chi tieu", "Gia tri"), rownames = FALSE,
        options = list(dom = "t", pageLength = 20, ordering = FALSE,
          initComplete = JS(
            "function(settings, json) {",
            "  $(this.api().table().header()).css({",
            "    'background-color':'#1F3864','color':'#FFFFFF',",
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif','font-weight':'600'",
            "  });",
            "  $(this.api().table().body()).css({",
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif'",
            "  });",
            "}"
          )
        )
      )
    })

    output$table_audit <- renderDT({
```

```
      df <- get_data()
      q3 <- quantile(df$PHIBH, 0.75, na.rm = T)
      iqr <- IQR(df$PHIBH, na.rm = T)
      upper <- q3 + 1.5 * iqr

      audit <- df %>%
        mutate(Flag = case_when(
          PHIBH > upper & XHKH %in% c("Kim cuong", "Bach kim") ~ "VIP + Outlier",
          PHIBH > upper ~ "Outlier Phi cao",
          PHIBH <= 1000 ~ "Phi cuc thap",
          TRUE ~ NA_character_
        )) %>%
        filter(!is.na(Flag)) %>%
        select(MA_HD, MA_KH, TENGOISANPHAM, TEN_CN, PHIBH, XHKH, Flag) %>%
        arrange(desc(PHIBH)) %>% head(200)

      datatable(audit, rownames = FALSE,
        options = list(scrollX = TRUE, pageLength = 10,
          initComplete = JS(
            "function(settings, json) {",
            "  $(this.api().table().header()).css({",
            "    'background-color':'#1F3864','color':'#FFFFFF',",
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif','font-weight':'600'",
            "  });",
            "  $(this.api().table().body()).css({",
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif'",
            "  });",
            "}"
          )
        )
      ) %>%
        formatCurrency("PHIBH", currency = "", interval = 3, mark = ".", digits = 0) %>%
        formatStyle("Flag",
          backgroundColor = styleEqual(
            c("VIP + Outlier", "Outlier Phi cao", "Phi cuc thap"),
            c("#D62728",       "#FF7F0E",         "#2CA02C")),
          color = "#FFFFFF",
          fontWeight = "bold")
    })
  })
}
```

Source: phase2.R

```
phase2UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 2: Khai pha vi mo va xu huong doanh thu",
             style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    p("Phan tich xu huong GWP theo thoi gian, co cau san pham va phan phoi phi bao hiem.",
                      style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),

    # --- KPI Cards ---
    fluidRow(
      valueBoxOutput(ns("vbox_total_gwp"), width = 3),
      valueBoxOutput(ns("vbox_avg_gwp"), width = 3),
      valueBoxOutput(ns("vbox_new_pct"), width = 3),
      valueBoxOutput(ns("vbox_renewal_pct"), width = 3)
    ),


    # --- Row 1: Xu huong GWP ---
    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Xu huong doanh thu GWP theo thang",
                                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Bien dong tong phi BH",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
```

```
        plotlyOutput(ns("plot_gwp_trend"), height = "380px"))
    ),

    # --- Row 2: Top SP + Density ---
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
        tags$h4("Top 10 goi san pham theo doanh thu",
                style = "color:#303983; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
        tags$p("",
               style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
        plotlyOutput(ns("plot_product"), height = "380px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
        tags$h4("Mat do phan phoi phi BH",
                style = "color:#303983; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
        tags$p("",
               style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
        plotlyOutput(ns("plot_density"), height = "380px"))
    ),

    # --- Row 3: Quarterly + New/Renewal ---
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
        tags$h4("Doanh thu theo quy",
                style = "color:#303983; font-weight:700; margin:0 0 2px 0;
```

```
font-family:'Helvetica Neue',sans-serif;"),

          tags$p("",

                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),

          plotlyOutput(ns("plot_qoq"), height = "380px")),

      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,

          tags$h4("Co cau HD va tai tuc theo thang",

                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),

          tags$p("So sanh doanh thu hop dong moi va tai tuc qua cac thang.",

                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),

          plotlyOutput(ns("plot_new_renewal"), height = "380px"))

    ),



    # --- Row 4 MOI: Doanh thu theo Nhom SP (multi-line) + So luong goi MoM ---

    fluidRow(

      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,

          tags$h4("Xu huong DT theo nhom SP",

                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),

          tags$p("",

                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),

          plotlyOutput(ns("plot_product_trend"), height = "380px")),

      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,

          tags$h4("So luong goi BH theo thang",

                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),

          tags$p("Bien dong phat hanh moi thang.",
```

```
            style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
        plotlyOutput(ns("plot_volume_trend"), height = "380px"))
    )
  )
}

phase2Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {

    get_data <- reactive({ req(all_data()); all_data()$master })

    # Font chuan plotly (giong PPTX)
    pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")

    # Layout helper  om sat box, grid nhat
    ly <- function(p, bl = 10, bb = 10, bt = 20, br = 10) {
      p %>% layout(
        font = pf,
        margin = list(l = bl, r = br, t = bt, b = bb),
        xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB", gridwidth = 0.5),
        yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB", gridwidth = 0.5)
      ) %>% config(displayModeBar = FALSE)
    }

    # ================================================================
```

```
# VALUE BOXES
# =================================================================
output$vbox_total_gwp <- renderValueBox({
  valueBox(format_vnd(sum(get_data()$PHIBH, na.rm = TRUE)), "Tong GWP",
           icon = icon("coins"), color = "blue")
})
output$vbox_avg_gwp <- renderValueBox({
  valueBox(format_vnd(mean(get_data()$PHIBH, na.rm = TRUE)), "GWP binh quan/goi",
           icon = icon("calculator"), color = "blue")
})
output$vbox_new_pct <- renderValueBox({
  df <- get_data()
  pct <- round(sum(df$LOAI_HD == "Moi", na.rm = TRUE) / nrow(df) * 100, 1)
  valueBox(paste0(pct, "%"), "Hop dong moi", icon = icon("plus-circle"), color = "teal")
})
output$vbox_renewal_pct <- renderValueBox({
  df <- get_data()
  pct <- round(sum(df$LOAI_HD == "Tai tuc", na.rm = TRUE) / nrow(df) * 100, 1)
  valueBox(paste0(pct, "%"), "Tai tuc", icon = icon("sync-alt"), color = "yellow")
})


output$plot_gwp_trend <- renderPlotly({
  df_sum <- get_data() %>%
    group_by(THANG) %>%
    summarise(GWP = sum(PHIBH, na.rm = TRUE), SL = n(), .groups = "drop") %>%
```

```
    arrange(THANG)

  p <- ggplot(df_sum, aes(x = THANG, y = GWP)) +
    geom_line(color = "#4A6FA5", linewidth = 1.2) +
    scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +
    scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0.02, 0.08))) +
    theme_minimal(base_family = "sans") +
    theme(
      panel.grid.major = element_line(color = "#EBEBEB", linewidth = 0.4),
      panel.grid.minor = element_blank(),
      axis.text.x = element_text(color = "#6C7A89", size = 9),
      axis.text.y = element_text(color = "#6C7A89"),
      axis.title = element_text(color = "#4A4A4A", face = "bold", size = 10),
      plot.margin = margin(0, 0, 0, 0)
    ) +
    labs(x = "Date", y = "Doanh thu GWP")

  ggplotly(p, tooltip = c("x", "y")) %>%
    style(line = list(shape = "spline", smoothing = 1.3)) %>%
    ly(bt = 15)
})


output$plot_product <- renderPlotly({
  df_prod <- get_data() %>%
```

```
    filter(!is.na(TENGOISANPHAM), TENGOISANPHAM != "Chua xac dinh") %>%

    group_by(TENGOISANPHAM) %>%

    summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%

    top_n(10, GWP)


  p <- ggplot(df_prod, aes(x = reorder(TENGOISANPHAM, GWP), y = GWP)) +

    geom_col(fill = "#4A6FA5", width = 0.65) +

    coord_flip() +

    scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +

    theme_minimal(base_family = "sans") +

    theme(

      panel.grid.major.y = element_blank(),

      panel.grid.major.x = element_line(color = "#EBEBEB", linewidth = 0.4),

      panel.grid.minor = element_blank(),

      axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),

      axis.title = element_text(color = "#4A4A4A", face = "bold", size = 10),

      plot.margin = margin(0, 0, 0, 0)

    ) +

    labs(x = "", y = "GWP (VND)")


  ggplotly(p) %>% ly(bl = 10, bb = 20)

})


output$plot_density <- renderPlotly({

  df <- get_data() %>% filter(PHIBH > 0)
```

```
  p <- ggplot(df, aes(x = PHIBH)) +
    geom_density(fill = "#2EC4B6", alpha = 0.6, color = "white", linewidth = 0.5) +
    scale_x_log10(labels = label_trieu) +
    scale_y_continuous(expand = expansion(mult = c(0, 0.08))) +
    theme_minimal(base_family = "sans") +
    theme(
      panel.grid.major = element_line(color = "#EBEBEB", linewidth = 0.4),
      panel.grid.minor = element_blank(),
      axis.title = element_text(color = "#4A4A4A", face = "bold", size = 10),
      axis.text = element_text(color = "#6C7A89"),
      plot.margin = margin(0, 0, 0, 0)
    ) +
    labs(x = "Phi BH (VND)  Log", y = "Mat do")

  ggplotly(p) %>% ly(bb = 25)
})


output$plot_qoq <- renderPlotly({
  df_q <- get_data() %>%
    mutate(Quy = paste0(year(NGAY_KY_HD), "-Q", quarter(NGAY_KY_HD))) %>%
    group_by(Quy) %>%
    summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%
    arrange(Quy)
```

```
    p <- ggplot(df_q, aes(x = Quy, y = GWP)) +
      geom_col(fill = "#55bded", width = 0.55) +
      scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
      theme_minimal(base_family = "sans") +
      theme(
          axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", size = 9, color =
"#4A4A4A"),
        panel.grid.major.x = element_blank(),
        panel.grid.major.y = element_line(color = "#EBEBEB", linewidth = 0.4),
        panel.grid.minor = element_blank(),
        axis.title = element_text(color = "#4A4A4A", face = "bold", size = 10),
        plot.margin = margin(0, 0, 0, 0)
      ) +
      labs(x = "", y = "GWP (VND)")

    ggplotly(p) %>% ly(bb = 30)
  })


  output$plot_new_renewal <- renderPlotly({
    df_nr <- get_data() %>%
      group_by(THANG, LOAI_HD) %>%
      summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop")
```

```
      p <- ggplot(df_nr, aes(x = THANG, y = GWP, fill = LOAI_HD)) +

        geom_area(alpha = 0.55, position = "stack") +

        scale_fill_manual(values = c("Moi" = "#4A6FA5", "Tai tuc" = "#FFCA28")) +

        scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +

        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.05))) +

        theme_minimal(base_family = "sans") +

        theme(

          panel.grid.major = element_line(color = "#EBEBEB", linewidth = 0.4),

          panel.grid.minor = element_blank(),

          legend.title = element_blank(),

          axis.title = element_text(color = "#4A4A4A", face = "bold", size = 10),

          axis.text = element_text(color = "#6C7A89"),

          plot.margin = margin(0, 0, 0, 0)

        ) +

        labs(x = "Date", y = "GWP (VND)")


      ggplotly(p) %>%

        layout(

          font = pf,

          margin = list(l = 10, r = 10, t = 40, b = 15),

          xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),

          yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),

            legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor = "bottom")

        ) %>% config(displayModeBar = FALSE)

    })
```

```r
output$plot_product_trend <- renderPlotly({

  df <- get_data()

  top5 <- df %>%

    filter(NHOMSANPHAM != "Chua xac dinh") %>%

    group_by(NHOMSANPHAM) %>% summarise(v = sum(PHIBH)) %>%

    top_n(5, v) %>% pull(NHOMSANPHAM)



  df_trend <- df %>%

    filter(NHOMSANPHAM %in% top5) %>%

    group_by(THANG, NHOMSANPHAM) %>%

    summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop")



  pal5 <- c("#7E57C2", "#2EC4B6", "#F06292", "#64B5F6", "#FFCA28")



  p <- ggplot(df_trend, aes(x = THANG, y = GWP, color = NHOMSANPHAM)) +

    geom_line(linewidth = 1.1) +

    scale_color_manual(values = pal5) +

    scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +

    scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0.02, 0.08))) +

    theme_minimal(base_family = "sans") +

    theme(

      panel.grid.major = element_line(color = "#EBEBEB", linewidth = 0.4),

      panel.grid.minor = element_blank(),
```

```
        legend.title = element_blank(),
        axis.title = element_text(color = "#4A4A4A", face = "bold", size = 10),
        axis.text = element_text(color = "#6C7A89"),
        plot.margin = margin(0, 0, 0, 0)
      ) +
      labs(x = "Date", y = "GWP (VND)")

    ggplotly(p) %>%
      style(line = list(shape = "spline", smoothing = 1.3)) %>%
      layout(
        font = pf,
        margin = list(l = 10, r = 10, t = 40, b = 15),
        xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),
        yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),
          legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom")
      ) %>% config(displayModeBar = FALSE)
  })

  output$plot_volume_trend <- renderPlotly({
    df_vol <- get_data() %>%
      group_by(THANG) %>%
      summarise(N = n(), .groups = "drop") %>%
      arrange(THANG)
```

```
      plot_ly(df_vol, x = ~THANG) %>%
        add_bars(y = ~N, name = "So luong goi",
                 marker = list(color = "#55bded", opacity = 0.6),
                 hovertemplate = "%{x|%b %Y}: %{y:,.0f} goi<extra></extra>") %>%
        add_lines(y = ~N, name = "Trend",
                  line = list(color = "#303983", width = 2, shape = "spline", smoothing = 1.3),
                  hoverinfo = "skip") %>%
        layout(
          font = pf,
          margin = list(l = 10, r = 10, t = 40, b = 15),
          xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB", gridwidth = 0.5,
                       tickformat = "%b %Y", dtick = "M3"),
          yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB", gridwidth = 0.5,
                           title = list(text = "So luong goi BH", font = list(size = 11, color =
"#4A4A4A")),
                       separatethousands = TRUE),
             legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom"),
          showlegend = TRUE,
          bargap = 0.3
        ) %>% config(displayModeBar = FALSE)
    })
  })
}
```

Source: phase3.R

```
phase3UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 3: Phan tich chuyen sau san pham va kenh ban",
         style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    p("Phan tich cau truc doanh thu, xac dinh dong san pham chu luc va hieu qua kenh phan phoi.",
         style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),

    fluidRow(
      valueBoxOutput(ns("vbox_avg_premium"), width = 4),
      valueBoxOutput(ns("vbox_top_product"), width = 4),
      valueBoxOutput(ns("vbox_top_channel"), width = 4)
    ),

    fluidRow(
      box(title = NULL, width = 8, status = "primary", solidHeader = FALSE,
          tags$h4("So sanh ty trong 2 mang theo chi nhanh",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Bieu do doi xung 2 ben voi thang do doc lap. Giup thay ro do manh yeu cua tung chi nhanh o ca mang doanh thu lon va nho.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_grouped_bar_matrix"), height = "480px")),
      box(title = NULL, width = 4, status = "primary", solidHeader = FALSE,
          tags$h4("Co cau doanh thu theo kenh ban",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Ty trong GWP cua tung kenh ban chinh.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_channel_pie"), height = "440px"))
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Phan phoi phi BH theo nhom san pham (Violin + Box)",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Xac dinh nguong phi pho bien va do phan tan cua tung nhom.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_premium_dist"), height = "390px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Hieu suat Top 15 kenh ban chi tiet",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Xep hang 15 kenh ban chi tiet co GWP cao nhat.",
```

```
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_detail_channel_bar"), height = "390px"))
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Top 10 goi san pham: Doanh thu & Ty trong tich luy",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Bar = GWP tung goi, nhan = % tich luy. Cho thay may goi tao ra phan lon doanh
thu.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_top10_cumulative"), height = "390px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Co cau nhom san pham theo kenh ban (Stacked %)",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Moi cot = 100%. Cho thay kenh nao tap trung vao nhom san pham nao.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_channel_product_stack"), height = "390px"))
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Phi binh quan/goi theo nhom san pham (APPP)",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Nhom nao co gia tri trung binh/goi cao nhat? Nhom nao volume lon nhung gia tri
thap?",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_appp_by_group"), height = "390px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("So luong goi BH va GWP theo nhom san pham",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Truc X = so goi, truc Y = phi binh quan, kich thuoc = tong GWP.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_bubble_product"), height = "390px"))
    ),

    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Bang tong hop hieu suat kinh doanh da chieu",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  6px  0;
font-family:'Helvetica Neue',sans-serif;"),
          DTOutput(ns("table_deep_stats")))
    )
  )
}
```

```
phase3Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {

    get_data <- reactive({ req(all_data()); all_data()$master })
    pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")
        pal  <-  c("#4A6FA5",  "#2EC4B6",  "#7E57C2",  "#F06292",  "#FFCA28",  "#55bded",  "#66BB6A",
"#EF5350", "#95A5A6")
    ly <- function(p, bl = 10, bb = 10, bt = 10, br = 10) {
      p %>% layout(font = pf, margin = list(l = bl, r = br, t = bt, b = bb),
                          xaxis = list(automargin = TRUE), yaxis = list(automargin = TRUE)) %>%
config(displayModeBar = FALSE)
    }

    output$vbox_avg_premium <- renderValueBox({
       valueBox(format_vnd(mean(get_data()$PHIBH, na.rm = TRUE)), "Phi binh quan/goi (APPP)", icon
= icon("dollar-sign"), color = "blue")
    })
    output$vbox_top_product <- renderValueBox({
      top <- get_data() %>% filter(TENGOISANPHAM != "Chua xac dinh") %>%
        group_by(TENGOISANPHAM) %>% summarise(v = sum(PHIBH)) %>% arrange(desc(v)) %>% slice(1)
      valueBox(top$TENGOISANPHAM, "SP doanh thu cao nhat", icon = icon("star"), color = "blue")
    })
    output$vbox_top_channel <- renderValueBox({
      top <- get_data() %>% filter(TENKENHBAN != "Chua xac dinh") %>%
        group_by(TENKENHBAN) %>% summarise(v = sum(PHIBH)) %>% arrange(desc(v)) %>% slice(1)
      valueBox(top$TENKENHBAN, "Kenh dan dau", icon = icon("trophy"), color = "teal")
    })

    output$plot_grouped_bar_matrix <- renderPlotly({
      # 1. Loc va tinh tong data
      df_matrix <- get_data() %>%
        filter(TEN_CN != "Chua xac dinh", TEN_CN != "Hoi so", NHOMSANPHAM != "Chua xac dinh") %>%
        group_by(TEN_CN, NHOMSANPHAM) %>%
        summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%
        arrange(desc(TEN_CN))

      # Chuyen doi du lieu ra don vi "Ty" de truc hien thi dep hon
      df_matrix <- df_matrix %>% mutate(GWP_Ty = GWP / 1e9)

      # 2. Tach data lam 2 nhanh
      df_bb <- df_matrix %>% filter(NHOMSANPHAM == "Bao hiem bat buoc")
      df_tn <- df_matrix %>% filter(NHOMSANPHAM == "Bao hiem tu nguyen")

      # 3. Canh Trai (Bat buoc)
      p1 <- plot_ly(df_bb, x = ~GWP_Ty, y = ~TEN_CN, type = 'bar', orientation = 'h',
                    name = "Bao hiem bat buoc", marker = list(color = "#4A6FA5"),
                    hovertemplate = "%{x:,.1f} ty<extra></extra>") %>%
         layout(xaxis = list(autorange = "reversed", title = "", gridcolor = "#EBEBEB", zeroline =
FALSE, ticksuffix = " ty"),
                   yaxis = list(side = "right", title = "", tickfont = list(family = "'Helvetica
```

```
Neue',sans-serif", size = 11, color = "#4A4A4A")))

      # 4. Canh Phai (Tu nguyen)
      p2 <- plot_ly(df_tn, x = ~GWP_Ty, y = ~TEN_CN, type = 'bar', orientation = 'h',
                    name = "Bao hiem tu nguyen", marker = list(color = "#2EC4B6"),
                    hovertemplate = "%{x:,.1f} ty<extra></extra>") %>%
          layout(xaxis = list(title = "", gridcolor = "#EBEBEB", zeroline = FALSE, ticksuffix = "
ty"),
               yaxis = list(showticklabels = FALSE))

      # 5. Ghep 2 bieu do (Ep margin nho lai con 0.08 de bieu do khit vao nhau)
      subplot(p1, p2, shareY = FALSE, titleX = FALSE, margin = 0.08) %>%
        layout(font = list(family = "'Helvetica Neue',sans-serif", color = "#4A4A4A"),
               margin = list(l = 10, r = 10, t = 40, b = 20),
               showlegend = TRUE,
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.15, yanchor =
"bottom", font = list(size = 10)),
               hovermode = "y unified") %>%
        config(displayModeBar = FALSE)
    })

    output$plot_channel_pie <- renderPlotly({
      df_chan <- get_data() %>% filter(TENKENHBAN != "Chua xac dinh") %>%
          group_by(TENKENHBAN) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%
arrange(desc(GWP))
         plot_ly(df_chan, labels = ~TENKENHBAN, values = ~GWP, type = "pie", textinfo = "percent",
hole = 0.45,
               marker = list(colors = pal[1:nrow(df_chan)], line = list(color = "#FFFFFF", width =
1.5)),
                    textfont = list(family = "'Helvetica Neue',sans-serif", size = 11, color =
"#FFFFFF")) %>%
        layout(font = pf, showlegend = TRUE,
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom", font = list(size = 10)),
               margin = list(l = 10, r = 10, t = 45, b = 10)) %>% config(displayModeBar = FALSE)
    })

    output$plot_premium_dist <- renderPlotly({
      df <- get_data() %>% filter(PHIBH > 0, NHOMSANPHAM != "Chua xac dinh")
      p <- ggplot(df, aes(x = NHOMSANPHAM, y = PHIBH, fill = NHOMSANPHAM)) +
        geom_violin(alpha = 0.65, scale = "width", color = NA) +
         geom_boxplot(width = 0.13, fill = "white", alpha = 0.85, color = "#303983", outlier.shape
= NA, linewidth = 0.4) +
        scale_y_log10(labels = label_trieu) + scale_fill_manual(values = pal) +
        theme_minimal(base_family = "sans") +
         theme(axis.text.x = element_text(angle = 22, hjust = 1, face = "bold", size = 9, color =
"#4A4A4A"),
               panel.grid.major.y = element_line(color = "#EBEBEB", linetype = "dashed", linewidth
= 0.35),
                    panel.grid.major.x = element_blank(), panel.grid.minor = element_blank(),
```

```
legend.position = "none",
              axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "Phi BH (VND)  Log")
      ggplotly(p) %>% ly(bl = 60, bb = 55)
    })

    output$plot_detail_channel_bar <- renderPlotly({
      df_det <- get_data() %>% filter(TENKENHBANCHITIET != "Chua xac dinh") %>%
           group_by(TENKENHBANCHITIET) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups =
"drop") %>% top_n(15, GWP)
      p <- ggplot(df_det, aes(x = reorder(TENKENHBANCHITIET, GWP), y = GWP)) +
        geom_col(fill = "#4A6FA5", width = 0.68) + coord_flip() +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
        theme_minimal(base_family = "sans") +
        theme(axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
              panel.grid.major.y = element_blank(),
              panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth
= 0.35),
              panel.grid.minor = element_blank(),
              axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "GWP (VND)")
      ggplotly(p) %>% ly(bl = 10, bb = 25)
    })

    output$plot_top10_cumulative <- renderPlotly({
      df_top <- get_data() %>%
        filter(TENGOISANPHAM != "Chua xac dinh", !is.na(TENGOISANPHAM)) %>%
          group_by(TENGOISANPHAM) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop")
%>%
        arrange(desc(GWP)) %>% head(10) %>%
        mutate(Cum_Pct = round(cumsum(GWP) / sum(get_data()$PHIBH, na.rm = TRUE) * 100, 1),
               Pct = round(GWP / sum(get_data()$PHIBH, na.rm = TRUE) * 100, 1))

      p <- ggplot(df_top, aes(x = reorder(TENGOISANPHAM, GWP), y = GWP)) +
        geom_col(fill = "#4A6FA5", width = 0.68) +
        geom_text(aes(label = paste0(Pct, "% | ", Cum_Pct, "%")),
                  hjust = -0.03, size = 3, color = "#303983", fontface = "bold") +
        coord_flip() +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.35))) +
        theme_minimal(base_family = "sans") +
        theme(axis.text.y = element_text(face = "bold", size = 8.5, color = "#4A4A4A"),
              panel.grid.major.y = element_blank(),
              panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth
= 0.35),
              panel.grid.minor = element_blank(),
              axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "GWP (VND)")
```

```
    ggplotly(p, tooltip = "y") %>% ly(bl = 10, bb = 25)
  })

  output$plot_channel_product_stack <- renderPlotly({
    df_stack <- get_data() %>%
      filter(TENKENHBAN != "Chua xac dinh", NHOMSANPHAM != "Chua xac dinh") %>%
      group_by(TENKENHBAN, NHOMSANPHAM) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups =
"drop")
    p <- ggplot(df_stack, aes(x = TENKENHBAN, y = GWP, fill = NHOMSANPHAM)) +
      geom_col(position = "fill", width = 0.65) +
        scale_y_continuous(labels = scales::percent_format(), expand = expansion(mult = c(0,
0.02))) +
      scale_fill_manual(values = pal) +
      theme_minimal(base_family = "sans") +
      theme(axis.text.x = element_text(angle = 30, hjust = 1, face = "bold", size = 9, color =
"#4A4A4A"),
            panel.grid.major.x = element_blank(),
            panel.grid.major.y = element_line(color = "#EBEBEB", linetype = "dashed", linewidth
= 0.35),
            panel.grid.minor = element_blank(), legend.title = element_blank(),
            axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
      labs(x = "", y = "Ty trong (%)")
    ggplotly(p) %>%
      layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 55),
             xaxis = list(automargin = TRUE), yaxis = list(automargin = TRUE),
             legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom", font = list(size = 9))
      ) %>% config(displayModeBar = FALSE)
  })

  output$plot_appp_by_group <- renderPlotly({
    df_appp <- get_data() %>%
      filter(NHOMSANPHAM != "Chua xac dinh") %>%
      group_by(NHOMSANPHAM) %>%
      summarise(APPP = mean(PHIBH, na.rm = TRUE), SL = n(), .groups = "drop")
    p <- ggplot(df_appp, aes(x = reorder(NHOMSANPHAM, APPP), y = APPP, fill = NHOMSANPHAM)) +
      geom_col(width = 0.62) +
       geom_text(aes(label = label_trieu(APPP)), hjust = -0.05, size = 3.2, color = "#303983",
fontface = "bold") +
      coord_flip() +
      scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.25))) +
      scale_fill_manual(values = pal) +
      theme_minimal(base_family = "sans") +
      theme(panel.grid.major.y = element_blank(),
            panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth
= 0.35),
            panel.grid.minor = element_blank(), legend.position = "none",
            axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
            axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
```

```
= margin(0,0,0,0)) +
        labs(x = "", y = "Phi binh quan/goi (VND)")
      ggplotly(p, tooltip = "y") %>% ly(bl = 10, bb = 25)
    })

    output$plot_bubble_product <- renderPlotly({
      df_bub <- get_data() %>%
        filter(NHOMSANPHAM != "Chua xac dinh") %>%
        group_by(NHOMSANPHAM) %>%
          summarise(Volume = n(), Revenue = sum(PHIBH, na.rm = TRUE), APPP = Revenue / Volume,
.groups = "drop")

       p <- ggplot(df_bub, aes(x = Volume, y = APPP, size = Revenue, color = NHOMSANPHAM, text =
NHOMSANPHAM)) +
        geom_point(alpha = 0.8) +
        scale_y_continuous(labels = label_trieu) +
        scale_x_continuous(labels = function(x) formatC(x, format = "d", big.mark = ".")) +
        scale_size_continuous(range = c(5, 22), guide = "none") +
        scale_color_manual(values = pal) +
        theme_minimal(base_family = "sans") +
        theme(panel.grid.major = element_line(color = "#EBEBEB", linewidth = 0.4),
              panel.grid.minor = element_blank(), legend.title = element_blank(),
              axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
              axis.text = element_text(color = "#6C7A89"), plot.margin = margin(0,0,0,0)) +
        labs(x = "So luong goi BH", y = "Phi binh quan (VND)")
      ggplotly(p, tooltip = c("text", "x", "y")) %>%
        layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 20),
               xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),
               yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom", font = list(size = 9))
        ) %>% config(displayModeBar = FALSE)
    })

    output$table_deep_stats <- renderDT({
      df_stat <- get_data() %>% filter(TEN_CN != "Chua xac dinh") %>%
        group_by(TEN_CN, NHOMSANPHAM, TENKENHBAN) %>%
          summarise(SL_Goi = n(), Tong_GWP = sum(PHIBH, na.rm = TRUE), APPP = round(Tong_GWP /
SL_Goi, 0), .groups = "drop") %>%
        arrange(desc(Tong_GWP))
      datatable(df_stat,
        colnames = c("Chi nhanh", "Nhom SP", "Kenh ban", "So goi", "Tong GWP", "Phi BQ/goi"),
        filter = "top", rownames = FALSE,
        options = list(pageLength = 10, scrollX = TRUE,
                                                                                  initComplete        =
JS("function(s,j){$(this.api().table().header()).css({'background-color':'#303983','color':'#fff',
'font-family':'Helvetica
Neue,sans-serif','font-weight':'600'});$(this.api().table().body()).css({'font-family':'Helvetica
Neue,sans-serif'});}"))) %>%
        formatCurrency(c("Tong_GWP", "APPP"), currency = "", interval = 3, mark = ".", digits = 0)
```

```
%>%
        formatRound("SL_Goi", digits = 0, interval = 3, mark = ".")
    })
  })
}
```

Source: phase4.R

```
library(shiny)
library(shinydashboard)
library(dplyr)
library(ggplot2)
library(plotly)
library(DT)
library(cluster) # Thu vien de tinh Silhouette
library(scales)  # Thu vien de Min-Max scaling

phase4UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
          h2("Phan 4: Phan cum khach hang RFM & K-Means",
             style = "color:#303983; font-weight:700; margin-bottom:4px;"),
             p("Phan tich phan khuc dua tren thuat toan K-Means ket hop kiem dinh Silhouette va
WSS.",
            style = "color:#6C7A89; margin-bottom:18px;"),

          # --- Control panel ---
          fluidRow(
            box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
                tags$h4("Cau hinh Thuat toan", style = "color:#303983; font-weight:700; margin:0 0
8px 0;"),
                fluidRow(
                   column(3, numericInput(ns("clusters"), "So cum (K):", value = 4, min = 2, max =
6)),
                  column(3, selectInput(ns("seed_val"), "Random Seed:", choices = c(42, 123, 456),
selected = 42)),
                  column(3, br(), actionButton(ns("run_kmeans"), "Chay Mo Hinh",
                                                  icon = icon("play-circle"), class = "btn-success
btn-lg")),
                  column(3, tags$p("Toi uu hoa toc do: Silhouette tinh tren tap mau n=10,000",
                                        style = "color:#E67E22; font-weight:bold; font-size:12px;
margin-top:12px;"))
                )
            )
          ),

          # --- Noi hien thi toan bo Dashboard ---
          uiOutput(ns("results_display"))
  )
}

phase4Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {
    ns <- session$ns
```

```
clustering_results <- reactiveVal(NULL)
diagnostics_data <- reactiveVal(NULL)

pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")
pal <- c("#4A6FA5", "#2EC4B6", "#7E57C2", "#F06292", "#FFCA28", "#55bded")

# Format layout chung cho Plotly
ly <- function(p, bl = 10, bb = 10, bt = 15, br = 10) {
  p %>% layout(
    font = pf, margin = list(l = bl, r = br, t = bt, b = bb),
    xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),
    yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB")
  ) %>% config(displayModeBar = FALSE)
}

observeEvent(input$run_kmeans, {
  withProgress(message = "Dang huan luyen mo hinh...", value = 0, {

    req(all_data()); df <- all_data()$master
    incProgress(0.2, detail = "Tinh toan RFM...")

    max_date <- max(df$NGAY_KY_HD, na.rm = TRUE)
    rfm_data <- df %>%
      group_by(MA_KH) %>%
      summarise(
        Recency   = as.numeric(max_date - max(NGAY_KY_HD, na.rm = TRUE)),
        Frequency = n(),
        Monetary  = sum(PHIBH, na.rm = TRUE),
        .groups = "drop"
      ) %>% filter(Monetary > 0, Frequency > 0)

    incProgress(0.4, detail = "Chuan hoa & Log-transform...")
    rfm_scaled <- rfm_data %>%
      mutate(R_log = log1p(Recency), F_log = log1p(Frequency), M_log = log1p(Monetary))
    scale_matrix <- scale(rfm_scaled[, c("R_log", "F_log", "M_log")])

    incProgress(0.5, detail = "Kiem dinh WSS & Silhouette (Sampling)...")
    # DUNG SAMPLING DE TRANH TRAN RAM KHI TINH SILHOUETTE CHO BIG DATA
    set.seed(42)
    samp_idx <- sample(1:nrow(scale_matrix), min(10000, nrow(scale_matrix)))
    samp_scale <- scale_matrix[samp_idx, ]
    dist_samp <- dist(samp_scale)

    wss <- numeric()
    sil_scores <- numeric()

    for(k in 2:8) {
      set.seed(as.numeric(input$seed_val))
      km_temp <- kmeans(scale_matrix, centers = k, nstart = 25)
      wss[k-1] <- km_temp$tot.withinss
```

```
          # Tinh Silhouette tren tap mau
          sil_temp <- cluster::silhouette(km_temp$cluster[samp_idx], dist_samp)
          sil_scores[k-1] <- mean(sil_temp[, 3])
        }
        diagnostics_data(data.frame(K = 2:8, WSS = wss, Silhouette = sil_scores))

        incProgress(0.8, detail = "Phan cum du lieu goc...")
        set.seed(as.numeric(input$seed_val))
        km <- kmeans(scale_matrix, centers = input$clusters, nstart = 25)
        rfm_data$Cluster <- as.factor(km$cluster)

        # Gan nhan dua tren Monetary
        clust_summary <- rfm_data %>% group_by(Cluster) %>% summarise(avg_m = mean(Monetary)) %>%
arrange(desc(avg_m))
        label_map <- setNames(c("VIP / Doanh nghiep", "KH Trung thanh", "KH Tiem nang", "KH Roi
bo", "Nhom 5", "Nhom 6")[1:input$clusters], clust_summary$Cluster)
        rfm_data$Cluster_Label <- label_map[as.character(rfm_data$Cluster)]

        attr(rfm_data, "km") <- km
        clustering_results(rfm_data)
        incProgress(1, detail = "Hoan tat!")
      })
    })

    # --- RENDER UI LAYOUT TONG THE ---
    output$results_display <- renderUI({
      req(clustering_results())
      tagList(
        # Row 1: Model Diagnostics (Elbow & Silhouette)
        fluidRow(
          box(title = NULL, width = 6, status = "primary",
              tags$h4("Elbow Method (WSS)", style = "color:#303983; font-weight:700;
font-size:14px;"),
              plotlyOutput(ns("plot_elbow"), height = "250px")),
          box(title = NULL, width = 6, status = "primary",
              tags$h4("Kiem dinh Silhouette", style = "color:#303983; font-weight:700;
font-size:14px;"),
              tags$p("Diem Silhouette cang gan 1, cac cum cang tach biet ro rang.", style =
"color:#6C7A89; font-size:11px; margin:0;"),
              plotlyOutput(ns("plot_silhouette"), height = "250px"))
        ),

        # Row 2: Bang Ho so Phan cum (Chinh) & Radar
        fluidRow(
          box(title = NULL, width = 7, status = "primary",
              tags$h4("Ho so Chan dung Khach hang (Cluster Profiling)", style = "color:#303983;
font-weight:700;"),
              DTOutput(ns("table_profiling"))),
          box(title = NULL, width = 5, status = "primary",
              tags$h4("Hinh thai R-F-M (Dai diem 0-100)", style = "color:#303983;
```

```
font-weight:700;"),
              plotlyOutput(ns("plot_radar"), height = "350px"))
        ),

        # Row 3: Scatter Plot & Bang Danh gia chat luong
        fluidRow(
          box(title = NULL, width = 8, status = "primary",
                    tags$h4("Phan tach Khong gian: Recency x Monetary", style = "color:#303983;
font-weight:700;"),
               tags$p("Hien thi dai dien mau 10,000 diem du lieu de tranh Overplotting.", style =
"color:#6C7A89; font-size:11px; margin:0;"),
              plotlyOutput(ns("plot_kmeans_rm"), height = "350px")),

          box(title = NULL, width = 4, status = "primary",
                       tags$h4("Danh gia mo hinh (Model Evaluation)", style = "color:#303983;
font-weight:700;"),
              uiOutput(ns("ui_quality_note")))
        )
      )
    })

    # --- CAC HAM VE BIEU DO ---

    # 1. Bieu do Elbow
    output$plot_elbow <- renderPlotly({
      edf <- diagnostics_data(); req(edf)
        p <- ggplot(edf, aes(x = K, y = WSS)) + geom_line(color = "#4A6FA5", linewidth = 1) +
geom_point(color = "#4A6FA5", size = 2) +
          geom_point(data = edf %>% filter(K == input$clusters), color = "#EF5350", size = 4) +
theme_minimal() + labs(x="K", y="WSS")
      ggplotly(p) %>% ly()
    })

    # 2. Bieu do Silhouette
    output$plot_silhouette <- renderPlotly({
      edf <- diagnostics_data(); req(edf)
      p <- ggplot(edf, aes(x = K, y = Silhouette)) + geom_line(color = "#2EC4B6", linewidth = 1) +
geom_point(color = "#2EC4B6", size = 2) +
          geom_point(data = edf %>% filter(K == input$clusters), color = "#EF5350", size = 4) +
theme_minimal() + labs(x="K", y="Silhouette Score")
      ggplotly(p) %>% ly()
    })

    # 3. Radar Chart (Truc Min-Max Scaling 0-100)
    output$plot_radar <- renderPlotly({
      res <- clustering_results(); req(res)

      radar_df <- res %>%
        group_by(Cluster_Label) %>%
          summarise(R_mean = mean(Recency), F_mean = mean(Frequency), M_mean = mean(Monetary),
```

```
.groups = "drop") %>%
        mutate(
          R_Score = rescale(R_mean, to = c(0, 100)),
          F_Score = rescale(F_mean, to = c(0, 100)),
          M_Score = rescale(M_mean, to = c(0, 100))
        )

      fig <- plot_ly(type = "scatterpolar", fill = "toself")
      for (i in 1:nrow(radar_df)) {
        fig <- fig %>% add_trace(
                         r = c(radar_df$R_Score[i], radar_df$F_Score[i], radar_df$M_Score[i],
radar_df$R_Score[i]),
          theta = c("Recency", "Frequency", "Monetary", "Recency"),
          name = radar_df$Cluster_Label[i],
          fillcolor = paste0(pal[i], "30"),
          line = list(color = pal[i], width = 2)
        )
      }

      fig %>% layout(
        polar = list(radialaxis = list(visible = TRUE, range = c(0, 100))),
        legend = list(orientation = "h", x = 0.5, xanchor = "center", y = -0.1)
      ) %>% config(displayModeBar = FALSE)
    })

    # 4. Data Table Tich hop In-cell Bar chart
    output$table_profiling <- renderDT({
      res <- clustering_results(); req(res)

      stats <- res %>%
        group_by(Cum = Cluster_Label) %>%
        summarise(
          So_KH = n(),
          Pct_Num = round(n() / nrow(res) * 100, 1),
          R_Days = round(mean(Recency), 0),
          F_Freq = round(mean(Frequency), 1),
          M_Value = sum(Monetary),
          .groups = "drop"
        ) %>%
        arrange(desc(M_Value)) %>%
        mutate(
          So_KH = formatC(So_KH, format = "d", big.mark = "."),
          M_Value = paste0(round(M_Value / 1e9, 1), " Ty")
        )

      datatable(stats,
                     colnames = c("Cum", "So luong", "Ty trong (%)", "R (Ngay)", "F (Tan suat)", "M
(Tong Chi)"),
                rownames = FALSE,
                options = list(dom = "t", ordering = FALSE, pageLength = 6)
```

```
      ) %>%
        formatStyle(
          'Pct_Num',
          background = styleColorBar(c(0, 100), '#E3F2FD'),
               backgroundSize = '98% 80%', backgroundRepeat = 'no-repeat', backgroundPosition =
'center'
        )
    })

    # 5. Scatter Plot (Voi ky thuat Sampling)
    output$plot_kmeans_rm <- renderPlotly({
      res <- clustering_results(); req(res)

      # Lay mau ngau nhien neu du lieu > 10.000 dong de bieu do muot va khong bi de mau
      res_plot <- if(nrow(res) > 10000) {
        set.seed(42); res %>% sample_n(10000)
      } else { res }

      p <- ggplot(res_plot, aes(x = Recency, y = Monetary, color = Cluster_Label)) +
        geom_point(alpha = 0.4, size = 1.2) +
        scale_y_continuous(labels = function(x) paste0(x / 1e6, " Tr")) +
        scale_color_manual(values = pal[1:input$clusters]) +
        theme_minimal() +
        theme(legend.title = element_blank()) +
        labs(x = "Recency (Ngay)", y = "Monetary (Trieu VND)")

         ggplotly(p) %>% ly() %>% layout(legend = list(orientation = "h", x = 0.5, xanchor =
"center", y = 1.1))
    })

    # 6. Bang Danh gia chat luong mo hinh
    output$ui_quality_note <- renderUI({
      res <- clustering_results(); req(res)
      km <- attr(res, "km")
      bss_pct <- round(km$betweenss / km$totss * 100, 1)

      tags$div(style = "font-family:'Helvetica Neue',sans-serif; color:#4A4A4A; font-size: 13px;",
               tags$table(style = "width:100%; border-collapse:collapse;",
                            tags$tr(tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;
font-weight:bold; width:45%;", "So cum (K)"),
                                   tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;",
input$clusters)),
                            tags$tr(tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;
font-weight:bold;", "Tong KH phan cum"),
                                   tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;",
formatC(nrow(res), format = "d", big.mark = "."))),
                            tags$tr(tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;
font-weight:bold;", "Between SS / Total SS"),
                                  tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;",
                                          paste0(bss_pct, "%"),
```

```
                                                    if(bss_pct >= 50) tags$span("  Tot", style =
"color:#66BB6A; font-weight:bold;")
                                           else tags$span("  Can xem xet", style = "color:#EF5350;
font-weight:bold;"))),
                            tags$tr(tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;
font-weight:bold;", "Preprocessing"),
                                  tags$td(style = "padding:8px; border-bottom:1px solid #EBEBEB;",
"log1p() -> scale(Z-score)")),
                                  tags$tr(tags$td(style = "padding:8px; font-weight:bold;", "Init
(nstart)"),
                                  tags$td(style = "padding:8px;", "25 (Tranh Local Optima)"))
               )
      )
    })

  })
}
```

Source: phase5.R

```
phase5UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 5: Phan tich mang luoi chi nhanh va hieu suat van hanh",
             style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    p("Danh gia suc khoe mang luoi, phan tich hieu suat chi nhanh va nhan vien, giam sat KPI.",
                      style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),

    # ===== SECTION 1: TONG QUAN CN =====
    tags$h3("1. Tong quan mang luoi chi nhanh",
                 style = "color:#4A6FA5; font-weight:700; font-size:17px; margin:10px 0 14px 0; font-family:'Helvetica Neue',sans-serif; border-bottom:2px solid #EBEBEB; padding-bottom:6px;"),
    fluidRow(
      box(title = NULL, width = 7, status = "primary", solidHeader = FALSE,
          tags$h4("Top chi nhanh theo doanh thu GWP",
                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Xep hang cac chi nhanh co tong GWP cao nhat.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_branch_gwp_bar"), height = "380px")),
      box(title = NULL, width = 5, status = "primary", solidHeader = FALSE,
          tags$h4("Co cau xep hang KH tai Top 3 CN",
                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Ty le Dong/Bac/Vang/Bach kim/Kim cuong tai 3 CN lon nhat.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_branch_xhkh"), height = "380px"))
    ),

    # ===== SECTION 2: XU HUONG =====
    tags$h3("2. Xu huong Doanh thu & Co cau San pham",
                 style = "color:#4A6FA5; font-weight:700; font-size:17px; margin:20px 0 14px 0; font-family:'Helvetica Neue',sans-serif; border-bottom:2px solid #EBEBEB; padding-bottom:6px;"),
    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Bien dong Doanh thu Top 5 CN theo Thang (Loai Hoi so)",
                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Multi-line chart: Top 5 CN co doanh thu cao nhat, loai tru Hoi so. Duong cong muot ma giup nhan dien xu huong ro hon.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_branch_trend_multi"), height = "380px"))
    ),
    fluidRow(
```

```
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Ma tran San pham  Chi nhanh (Top 10 CN)",
                                    style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Heatmap: Doanh thu GWP theo nhom san pham tai 10 CN co doanh thu cao nhat. Mau sac dam nhat the hien muc do dong gop cua tung nhom SP tai moi CN.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_matrix_prod_branch"), height = "440px"))
    ),

    # ===== SECTION 3: KPI =====
    tags$h3("3. Quan tri muc tieu (KPI)",
                 style = "color:#4A6FA5; font-weight:700; font-size:17px; margin:20px 0 14px 0; font-family:'Helvetica Neue',sans-serif; border-bottom:2px solid #EBEBEB; padding-bottom:6px;"),
    fluidRow(
      box(title = NULL, width = 9, status = "primary", solidHeader = FALSE,
          tags$h4("Giam sat doanh thu: Thuc te vs muc tieu (KPI)",
                                    style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Combo chart: Cot the hien doanh thu thuc te theo thang, duong net dut the hien muc tieu KPI. Cho phep chon loc theo kenh ban de danh gia hieu qua tung kenh.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_kpi_combo"), height = "420px")),
      box(title = NULL, width = 3, status = "primary", solidHeader = FALSE,
                 tags$h4("Bo loc", style = "color:#303983; font-weight:700; margin:0 0 8px 0; font-family:'Helvetica Neue',sans-serif;"),
          selectInput(ns("select_kpi_channel"), "Chon Kenh:", choices = "Tat ca"),
          hr(),
          valueBoxOutput(ns("vbox_kpi_pct"), width = 12))
    ),

    # ===== SECTION 4: NHAN VIEN =====
    tags$h3("4. Phan tich nang suat nhan vien",
                 style = "color:#4A6FA5; font-weight:700; font-size:17px; margin:20px 0 14px 0; font-family:'Helvetica Neue',sans-serif; border-bottom:2px solid #EBEBEB; padding-bottom:6px;"),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Phan phoi nang suat NV",
                                    style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Histogram the hien so luong nhan vien theo cac muc nang suat GWP, su dung thang log de hien thi ro hon su phan hoa giua cac nhan vien co nang suat thap va cao.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_staff_density"), height = "370px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("So sanh GWP theo kenh ban",
                                    style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Tong doanh thu GWP cua tung kenh ban chinh.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
```

```
          plotlyOutput(ns("plot_channel_comp"), height = "370px"))
    ),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Nang suat Trung binh/NV theo Chi Nhanh (Top 15)",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("GWP trung binh moi nhan vien tai tung chi nhanh.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_productivity_cn"), height = "370px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Bang xep hang Top 15 nhan vien xuat sac",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  6px  0;
font-family:'Helvetica Neue',sans-serif;"),
          DTOutput(ns("table_top_staff")))
    )
  )
}

phase5Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {

    get_data <- reactive({ req(all_data()); all_data()$master })
    get_kpi  <- reactive({ req(all_data()); all_data()$kpi })

    observe({
      kb <- get_data() %>% filter(TENKENHBAN != "Chua xac dinh") %>% pull(TENKENHBAN) %>% unique()
%>% sort()
      updateSelectInput(session, "select_kpi_channel", choices = c("Tat ca", kb))
    })

    pf  <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")
    pal <- c("#4A6FA5", "#2EC4B6", "#7E57C2", "#F06292", "#FFCA28", "#55bded", "#66BB6A")

    ly <- function(p, bl = 10, bb = 10, bt = 10, br = 10) {
      p %>% layout(
        font = pf, margin = list(l = bl, r = br, t = bt, b = bb),
        xaxis = list(automargin = TRUE), yaxis = list(automargin = TRUE)
      ) %>% config(displayModeBar = FALSE)
    }

    output$plot_branch_gwp_bar <- renderPlotly({
      df_cn <- get_data() %>%
        filter(TEN_CN != "Chua xac dinh") %>%
        group_by(TEN_CN) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%
        top_n(15, GWP)

      p <- ggplot(df_cn, aes(x = reorder(TEN_CN, GWP), y = GWP)) +
        geom_col(fill = "#4A6FA5", width = 0.68) + coord_flip() +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
```

```
        theme_minimal(base_family = "sans") +
        theme(
          panel.grid.major.y = element_blank(),
          panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
          panel.grid.minor = element_blank(),
          axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
          axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
          plot.margin = margin(0, 0, 0, 0)
        ) + labs(x = "", y = "GWP (VND)")

      ggplotly(p) %>% ly(bb = 25)
    })


    output$plot_branch_xhkh <- renderPlotly({
      df <- get_data()
      top3 <- df %>% filter(TEN_CN != "Chua xac dinh") %>%
        group_by(TEN_CN) %>% summarise(v = sum(PHIBH)) %>% top_n(3, v) %>% pull(TEN_CN)

      df_xh <- df %>% filter(TEN_CN %in% top3) %>%
        group_by(TEN_CN, XHKH) %>% summarise(N = n_distinct(MA_KH), .groups = "drop")

      p <- ggplot(df_xh, aes(x = TEN_CN, y = N, fill = XHKH)) +
        geom_col(position = "fill", width = 0.6) +
        scale_fill_manual(values = pal) +
        scale_y_continuous(labels = scales::percent_format(), expand = expansion(mult = c(0,
0.02))) +
        theme_minimal(base_family = "sans") +
        theme(
          panel.grid.major.x = element_blank(),
          panel.grid.major.y = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
          panel.grid.minor = element_blank(),
          axis.text.x = element_text(face = "bold", size = 9, color = "#4A4A4A"),
          axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
          legend.title = element_blank(),
          plot.margin = margin(0, 0, 0, 0)
        ) + labs(x = "", y = "Ty le")

      ggplotly(p) %>%
        layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 20),
               xaxis = list(automargin = TRUE), yaxis = list(automargin = TRUE),
               legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom",
                             font = list(size = 9))
        ) %>% config(displayModeBar = FALSE)
    })
```

```
    output$plot_branch_trend_multi <- renderPlotly({
      df <- get_data()
      top5 <- df %>%
        filter(TEN_CN != "Chua xac dinh", !grepl("Hoi so|Hoi so", TEN_CN, ignore.case = TRUE)) %>%
        group_by(TEN_CN) %>% summarise(v = sum(PHIBH)) %>% top_n(5, v) %>% pull(TEN_CN)

      df_trend <- df %>% filter(TEN_CN %in% top5) %>%
        group_by(THANG, TEN_CN) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop")

      p <- ggplot(df_trend, aes(x = THANG, y = GWP, color = TEN_CN)) +
        geom_line(linewidth = 1.15) +
        scale_color_manual(values = pal[1:5]) +
        scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0.02, 0.08))) +
        theme_minimal(base_family = "sans") +
        theme(
          panel.grid.major = element_line(color = "#EBEBEB", linewidth = 0.4),
          panel.grid.minor = element_blank(),
          legend.title = element_blank(),
          axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
          axis.text = element_text(color = "#6C7A89"),
          plot.margin = margin(0, 0, 0, 0)
        ) + labs(x = "Date", y = "GWP (VND)")

      ggplotly(p) %>%
        style(line = list(shape = "spline", smoothing = 1.3)) %>%
        layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 15),
               xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),
               yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB"),
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom",
                             font = list(size = 10))
        ) %>% config(displayModeBar = FALSE)
    })


    output$plot_matrix_prod_branch <- renderPlotly({
      df <- get_data()
      top10 <- df %>% filter(TEN_CN != "Chua xac dinh") %>%
        group_by(TEN_CN) %>% summarise(v = sum(PHIBH)) %>% top_n(10, v) %>% pull(TEN_CN)

      df_mat <- df %>%
        filter(TEN_CN %in% top10, NHOMSANPHAM != "Chua xac dinh") %>%
          group_by(TEN_CN, NHOMSANPHAM) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups =
"drop")

      p <- ggplot(df_mat, aes(x = TEN_CN, y = NHOMSANPHAM, fill = GWP)) +
        geom_tile(color = "white", linewidth = 0.6) +
        scale_fill_gradient(low = "#f0f2f7", high = "#303983", labels = label_trieu) +
        theme_minimal(base_family = "sans") +
```

```
        theme(
            axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", size = 9, color =
"#4A4A4A"),
          axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
          panel.grid = element_blank(), axis.title = element_blank(),
          plot.margin = margin(0, 0, 0, 0)
        )

      ggplotly(p) %>% ly(bl = 10, bb = 65)
    })


    output$plot_kpi_combo <- renderPlotly({
      df_act <- get_data(); df_kpi <- get_kpi()

      if (input$select_kpi_channel != "Tat ca") {
        kb_code <- df_act %>% filter(TENKENHBAN == input$select_kpi_channel) %>%
          pull(MA_KENHBAN) %>% unique() %>% first()
        df_act <- df_act %>% filter(TENKENHBAN == input$select_kpi_channel)
        df_kpi <- df_kpi %>% filter(MA_KENHBAN == kb_code)
      }

      act_sum <- df_act %>% mutate(YM = format(NGAY_KY_HD, "%Y%m")) %>%
        group_by(YM) %>% summarise(Actual = sum(PHIBH, na.rm = TRUE), .groups = "drop")
      kpi_sum <- df_kpi %>% group_by(YM = as.character(THANGNAM)) %>%
        summarise(Target = sum(DOANH_THU, na.rm = TRUE), .groups = "drop")

      df_combo <- full_join(act_sum, kpi_sum, by = "YM") %>% arrange(YM) %>%
        mutate(Label = paste0(substr(YM, 5, 6), "/", substr(YM, 1, 4)))

      plot_ly(df_combo, x = ~Label) %>%
        add_bars(y = ~Actual, name = "Thuc Te", marker = list(color = "#4A6FA5")) %>%
        add_lines(y = ~Target, name = "Muc Tieu (KPI)",
                  line = list(color = "#F06292", width = 3, dash = "dash", shape = "spline")) %>%
        layout(
          font = pf,
          yaxis = list(title = list(text = "Doanh Thu (VND)", font = list(size = 11)),
                       tickformat = ",.0f", gridcolor = "#EBEBEB"),
          xaxis = list(title = "", tickangle = -45, gridcolor = "#EBEBEB"),
          barmode = "group", bargap = 0.3,
             legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom"),
          margin = list(l = 10, r = 10, t = 40, b = 45)
        ) %>% config(displayModeBar = FALSE)
    })

    output$vbox_kpi_pct <- renderValueBox({
      total_act <- sum(get_data()$PHIBH, na.rm = TRUE)
      total_target <- sum(get_kpi()$DOANH_THU, na.rm = TRUE)
      pct <- ifelse(total_target > 0, round(total_act / total_target * 100, 1), 0)
```

```
    valueBox(paste0(pct, "%"), "Tien do KPI", icon = icon("check-double"),
             color = ifelse(pct >= 100, "green", ifelse(pct >= 80, "yellow", "red")))
  })


  output$plot_staff_density <- renderPlotly({
    df_staff <- get_data() %>%
      group_by(MA_NV) %>% summarise(Total_GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%
      filter(Total_GWP > 0)

    p <- ggplot(df_staff, aes(x = Total_GWP)) +
      geom_histogram(bins = 45, fill = "#2EC4B6", alpha = 0.8, color = "white", linewidth = 0.2)
+
      scale_x_log10(labels = label_trieu) +
      scale_y_continuous(expand = expansion(mult = c(0, 0.08))) +
      theme_minimal(base_family = "sans") +
      theme(
           panel.grid.major = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
        panel.grid.minor = element_blank(),
        axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
        axis.text = element_text(color = "#6C7A89"),
        plot.margin = margin(0, 0, 0, 0)
      ) + labs(x = "Tong GWP/NV (VND)  Log", y = "So NV")

    ggplotly(p) %>% ly(bb = 25)
  })


  output$plot_channel_comp <- renderPlotly({
    df_chan <- get_data() %>% filter(TENKENHBAN != "Chua xac dinh") %>%
      group_by(TENKENHBAN) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop")

    p <- ggplot(df_chan, aes(x = reorder(TENKENHBAN, GWP), y = GWP)) +
      geom_col(fill = "#4A6FA5", width = 0.62) + coord_flip() +
      scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
      theme_minimal(base_family = "sans") +
      theme(
        panel.grid.major.y = element_blank(),
          panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
        panel.grid.minor = element_blank(),
        axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
        axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
        plot.margin = margin(0, 0, 0, 0)
      ) + labs(x = "", y = "GWP (VND)")

    ggplotly(p) %>% ly(bb = 25)
  })
```

```
  output$plot_productivity_cn <- renderPlotly({
    df_prod <- get_data() %>%
      filter(TEN_CN != "Chua xac dinh", !grepl("Hoi so|Hoi so|TCT", TEN_CN, ignore.case = TRUE))
%>%
      group_by(TEN_CN, MA_NV) %>%
      summarise(GWP_NV = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%
      group_by(TEN_CN) %>%
      summarise(Avg_GWP_NV = mean(GWP_NV), N_NV = n(), .groups = "drop") %>%
      top_n(15, Avg_GWP_NV)

    p <- ggplot(df_prod, aes(x = reorder(TEN_CN, Avg_GWP_NV), y = Avg_GWP_NV)) +
      geom_col(fill = "#FFCA28", width = 0.65) + coord_flip() +
      scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
      theme_minimal(base_family = "sans") +
      theme(
        panel.grid.major.y = element_blank(),
          panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
        panel.grid.minor = element_blank(),
        axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
        axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
        plot.margin = margin(0, 0, 0, 0)
      ) + labs(x = "", y = "GWP TB/NV (VND)")

    ggplotly(p) %>% ly(bb = 25)
  })


  output$table_top_staff <- renderDT({
    df <- get_data() %>%
      filter(!grepl("Hoi so|Hoi so|TCT", TEN_CN, ignore.case = TRUE))

    # Tim kenh chinh cua moi NV (kenh co GWP cao nhat)
    kenh_chinh <- df %>%
      group_by(MA_NV, TENKENHBAN) %>%
      summarise(gwp_kenh = sum(PHIBH, na.rm = TRUE), .groups = "drop") %>%
      group_by(MA_NV) %>%
      slice_max(gwp_kenh, n = 1, with_ties = FALSE) %>%
      select(MA_NV, Kenh_Chinh = TENKENHBAN)

    df_top <- df %>%
      group_by(MA_NV, TEN_CN) %>%
      summarise(GWP = sum(PHIBH, na.rm = TRUE), SL_Goi = n(),
                So_Kenh = n_distinct(TENKENHBAN), .groups = "drop") %>%
      left_join(kenh_chinh, by = "MA_NV") %>%
      arrange(desc(GWP)) %>% head(15)

    datatable(df_top,
      colnames = c("Ma NV", "Chi nhanh", "Tong GWP", "So goi", "So kenh", "Kenh chinh"),
```

```
        rownames = FALSE,
        options = list(dom = "t", pageLength = 15,
          initComplete = JS(
            "function(settings, json) {",
            "  $(this.api().table().header()).css({",
            "    'background-color':'#303983','color':'#fff',",
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif','font-weight':'600'",
            "  });",
            "  $(this.api().table().body()).css({",
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif'",
            "  });",
            "}"
          )
        )
      ) %>%
        formatCurrency("GWP", currency = "", interval = 3, mark = ".", digits = 0) %>%
        formatRound("SL_Goi", digits = 0, interval = 3, mark = ".")
    })
  })
}
```

Source: phase6.R

```
phase6UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 6: Phan tich khach hang tiem nang va Du bao nhu cau",
       style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    p("Phan tich toan bo 400 nghin khach hang: ai da mua, ai chua mua, va co hoi chuyen doi o dau.",
      style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    fluidRow(
      valueBoxOutput(ns("vb_total_kh_all"), width = 3), valueBoxOutput(ns("vb_kh_bought"), width = 3),
      valueBoxOutput(ns("vb_kh_not_bought"), width = 3), valueBoxOutput(ns("vb_conversion_rate"), width = 3)
    ),
    fluidRow(
      box(title = NULL, width = 5, status = "primary", solidHeader = FALSE,
          tags$h4("Ty le da mua vs chua mua", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Trong toan bo KH he thong, bao nhieu % da phat sinh giao dich?", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_bought_vs_not"), height = "360px")),
      box(title = NULL, width = 7, status = "primary", solidHeader = FALSE,
          tags$h4("Phan bo xep hang: Da mua vs Chua mua", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Hang nao con nhieu KH chua duoc khai thac?", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_xhkh_compare"), height = "360px"))
    ),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Nhom tuoi KH chua mua", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Nhom tuoi nao co nhieu KH tiem nang nhat?", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_age_not_bought"), height = "370px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Nghe nghiep KH chua mua (Top 15)", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Nghe nao co tep KH tiem nang lon nhat?", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_job_not_bought"), height = "370px"))
    ),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Ty le chuyen doi theo xep hang", style = "color:#303983; font-weight:700;
```

```
margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Hang nao co ty le mua cao nhat?", style = "color:#6C7A89; font-size:12px;
margin-bottom:6px;"),
          plotlyOutput(ns("plot_conversion_by_xhkh"), height = "370px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Doanh thu tiem nang neu chuyen doi KH chua mua", style = "color:#303983;
font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Uoc tinh: KH chua mua  chi tieu TB hang tuong ung.", style = "color:#6C7A89;
font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_potential_revenue"), height = "370px"))
    ),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Gioi tinh KH chua mua", style = "color:#303983; font-weight:700; margin:0 0 2px
0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Ty le nam/nu trong tep KH tiem nang.", style = "color:#6C7A89; font-size:12px;
margin-bottom:6px;"),
          plotlyOutput(ns("plot_gender_not_bought"), height = "360px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Ty le chuyen doi theo nhom tuoi", style = "color:#303983; font-weight:700;
margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Nhom tuoi nao de chuyen doi nhat?", style = "color:#6C7A89; font-size:12px;
margin-bottom:6px;"),
          plotlyOutput(ns("plot_conversion_by_age"), height = "360px"))
    ),
    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Bang tong hop tiem nang theo xep hang", style = "color:#303983;
font-weight:700; margin:0 0 6px 0; font-family:'Helvetica Neue',sans-serif;"),
          DTOutput(ns("table_potential_summary")))
    )
  )
}

phase6Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {
    get_master <- reactive({ req(all_data()); all_data()$master })
    all_kh <- reactive({ req(file.exists("kh_full.rds")); readRDS("kh_full.rds") })
    kh_tagged <- reactive({
      bought_ids <- unique(get_master()$MA_KH)
      all_kh() %>% mutate(Status = ifelse(MA_KH %in% bought_ids, "Da mua", "Chua mua"))
    })
    avg_spend <- reactive({
      get_master() %>% group_by(XHKH, MA_KH) %>% summarise(T = sum(PHIBH, na.rm = TRUE), .groups =
"drop") %>%
        group_by(XHKH) %>% summarise(Avg = mean(T), .groups = "drop")
    })
    pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")

      output$vb_total_kh_all <- renderValueBox({ valueBox(formatC(nrow(all_kh()), format = "d",
```

```
big.mark = "."), "Tong KH he thong", color = "blue") })
    output$vb_kh_bought <- renderValueBox({ valueBox(formatC(sum(kh_tagged()$Status == "Da mua"), format = "d", big.mark = "."), "KH da mua", color = "teal") })
    output$vb_kh_not_bought <- renderValueBox({ valueBox(formatC(sum(kh_tagged()$Status == "Chua mua"), format = "d", big.mark = "."), "KH chua mua", color = "yellow") })
    output$vb_conversion_rate <- renderValueBox({
      pct <- round(sum(kh_tagged()$Status == "Da mua") / nrow(kh_tagged()) * 100, 1)
      valueBox(paste0(pct, "%"), "Ty le chuyen doi", color = ifelse(pct >= 50, "green", "red"))
    })

    output$plot_bought_vs_not <- renderPlotly({
      df <- kh_tagged() %>% group_by(Status) %>% summarise(N = n(), .groups = "drop")
      plot_ly(df, labels = ~Status, values = ~N, type = "pie", hole = 0.5, textinfo = "percent+value",
              marker = list(colors = c("#4A6FA5","#FFCA28"), line = list(color = "#FFF", width = 1.5)),
              textfont = list(size = 12, color = "#303983", family = "Helvetica Neue")) %>%
        layout(font = pf, showlegend = TRUE, legend = list(orientation = "h", x = 0.5, xanchor = "center", y = -0.05),
               margin = list(l = 10, r = 10, t = 10, b = 30)) %>% config(displayModeBar = FALSE)
    })

    output$plot_xhkh_compare <- renderPlotly({
      df <- kh_tagged() %>% group_by(XHKH, Status) %>% summarise(N = n(), .groups = "drop") %>%
        mutate(XHKH = factor(XHKH, levels = c("Dong","Bac","Vang","Bach kim","Kim cuong")))
      p <- ggplot(df, aes(x = XHKH, y = N, fill = Status)) + geom_col(position = "dodge", width = 0.7) +
        scale_fill_manual(values = c("Da mua"="#4A6FA5","Chua mua"="#FFCA28")) +
        scale_y_continuous(labels = function(x) formatC(x, format = "d", big.mark = "."), expand = expansion(mult = c(0, 0.08))) +
        theme_minimal(base_family = "sans") +
        theme(panel.grid.major.x = element_blank(), panel.grid.major.y = element_line(color = "#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), legend.title = element_blank(),
              axis.text.x = element_text(face = "bold", size = 10, color = "#4A4A4A"),
              axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin = margin(0,0,0,0)) + labs(x = "", y = "So KH")
      ggplotly(p) %>% layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 20),
               legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor = "bottom")) %>% config(displayModeBar = FALSE)
    })

    output$plot_age_not_bought <- renderPlotly({
      df <- kh_tagged() %>% filter(Status == "Chua mua", !is.na(NHOM_TUOI)) %>% group_by(NHOM_TUOI) %>% summarise(N = n(), .groups = "drop")
      p <- ggplot(df, aes(x = reorder(NHOM_TUOI, N), y = N)) + geom_col(fill = "#7E57C2", width = 0.62) +
        geom_text(aes(label = formatC(N, format = "d", big.mark = ".")), hjust = -0.05, size = 3.2, color = "#303983", fontface = "bold") +
        coord_flip() + scale_y_continuous(expand = expansion(mult = c(0, 0.22)), labels =
```

```
function(x) formatC(x, format = "d", big.mark = ".")) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.y = element_blank(), panel.grid.major.x = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), axis.text.y = element_text(face = "bold", size =
9, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) + labs(x = "", y = "So KH chua mua")
      ggplotly(p, tooltip = "y") %>% layout(font = pf, margin = list(l = 10, r = 20, t = 10, b =
25),
                                                         xaxis = list(automargin = TRUE), yaxis =
list(automargin = TRUE)) %>% config(displayModeBar = FALSE)
    })

    output$plot_job_not_bought <- renderPlotly({
      df <- kh_tagged() %>% filter(Status == "Chua mua") %>% group_by(NGHENGHIEP) %>% summarise(N
= n(), .groups = "drop") %>% top_n(15, N)
      p <- ggplot(df, aes(x = reorder(NGHENGHIEP, N), y = N)) + geom_col(fill = "#2EC4B6", width =
0.62) +
        geom_text(aes(label = formatC(N, format = "d", big.mark = ".")), hjust = -0.05, size = 3,
color = "#303983", fontface = "bold") +
            coord_flip() + scale_y_continuous(expand = expansion(mult = c(0, 0.22)), labels =
function(x) formatC(x, format = "d", big.mark = ".")) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.y = element_blank(), panel.grid.major.x = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), axis.text.y = element_text(face = "bold", size =
8.5, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) + labs(x = "", y = "So KH chua mua")
      ggplotly(p, tooltip = "y") %>% layout(font = pf, margin = list(l = 10, r = 20, t = 10, b =
25),
                                                         xaxis = list(automargin = TRUE), yaxis =
list(automargin = TRUE)) %>% config(displayModeBar = FALSE)
    })

    output$plot_conversion_by_xhkh <- renderPlotly({
      df <- kh_tagged() %>% group_by(XHKH) %>%
         summarise(Total = n(), Pct = round(sum(Status == "Da mua") / Total * 100, 1), .groups =
"drop") %>%
        mutate(XHKH = factor(XHKH, levels = c("Dong","Bac","Vang","Bach kim","Kim cuong")))
      p <- ggplot(df, aes(x = XHKH, y = Pct, fill = XHKH)) + geom_col(width = 0.6) +
          geom_text(aes(label = paste0(Pct, "%")), vjust = -0.5, size = 3.5, color = "#303983",
fontface = "bold") +
             scale_fill_manual(values = c("Dong"="#95A5A6","Bac"="#BDC3C7","Vang"="#FFCA28","Bach
kim"="#7E57C2","Kim cuong"="#303983")) +
        scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.x = element_blank(), panel.grid.major.y = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
```

```
            panel.grid.minor = element_blank(), legend.position = "none",
            axis.text.x = element_text(face = "bold", size = 10, color = "#4A4A4A"),
             axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) + labs(x = "", y = "Ty le chuyen doi (%)")
      ggplotly(p, tooltip = "y") %>% layout(font = pf, margin = list(l = 10, r = 10, t = 10, b =
25),
                                                         xaxis = list(automargin = TRUE), yaxis =
list(automargin = TRUE)) %>% config(displayModeBar = FALSE)
    })

    output$plot_potential_revenue <- renderPlotly({
       not_b <- kh_tagged() %>% filter(Status == "Chua mua") %>% group_by(XHKH) %>% summarise(N =
n(), .groups = "drop")
      df <- left_join(not_b, avg_spend(), by = "XHKH") %>% mutate(Pot = N * Avg,
           XHKH = factor(XHKH, levels = c("Dong","Bac","Vang","Bach kim","Kim cuong")))
      p <- ggplot(df, aes(x = XHKH, y = Pot, fill = XHKH)) + geom_col(width = 0.6) +
          geom_text(aes(label = label_trieu(Pot)), vjust = -0.5, size = 3.2, color = "#303983",
fontface = "bold") +
             scale_fill_manual(values = c("Dong"="#95A5A6","Bac"="#BDC3C7","Vang"="#FFCA28","Bach
kim"="#7E57C2","Kim cuong"="#303983")) +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.15))) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.x = element_blank(), panel.grid.major.y = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
            panel.grid.minor = element_blank(), legend.position = "none",
            axis.text.x = element_text(face = "bold", size = 10, color = "#4A4A4A"),
             axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) + labs(x = "", y = "DT tiem nang (VND)")
      ggplotly(p, tooltip = "y") %>% layout(font = pf, margin = list(l = 10, r = 10, t = 10, b =
25),
                                                         xaxis = list(automargin = TRUE), yaxis =
list(automargin = TRUE)) %>% config(displayModeBar = FALSE)
    })

    output$plot_gender_not_bought <- renderPlotly({
      df <- kh_tagged() %>% filter(Status == "Chua mua") %>% group_by(GIOI_TINH) %>% summarise(N =
n(), .groups = "drop")
         plot_ly(df, labels = ~GIOI_TINH, values = ~N, type = "pie", hole = 0.45, textinfo =
"percent+label",
              marker = list(colors = c("#4A6FA5","#F06292","#95A5A6"), line = list(color = "#FFF",
width = 1.5)),
            textfont = list(size = 12, color = "#303983", family = "Helvetica Neue")) %>%
         layout(font = pf, showlegend = FALSE, margin = list(l = 10, r = 10, t = 10, b = 10)) %>%
config(displayModeBar = FALSE)
    })

    output$plot_conversion_by_age <- renderPlotly({
      df <- kh_tagged() %>% filter(!is.na(NHOM_TUOI)) %>% group_by(NHOM_TUOI) %>%
         summarise(Total = n(), Pct = round(sum(Status == "Da mua") / Total * 100, 1), .groups =
"drop")
```

```
        p <- ggplot(df, aes(x = reorder(NHOM_TUOI, Pct), y = Pct)) + geom_col(fill = "#55bded",
width = 0.62) +
          geom_text(aes(label = paste0(Pct, "%")), hjust = -0.08, size = 3.3, color = "#303983",
fontface = "bold") +
        coord_flip() + scale_y_continuous(expand = expansion(mult = c(0, 0.18))) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.y = element_blank(), panel.grid.major.x = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), axis.text.y = element_text(face = "bold", size =
9, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) + labs(x = "", y = "Ty le chuyen doi (%)")
      ggplotly(p, tooltip = "y") %>% layout(font = pf, margin = list(l = 10, r = 20, t = 10, b =
25),
                                                         xaxis = list(automargin = TRUE), yaxis =
list(automargin = TRUE)) %>% config(displayModeBar = FALSE)
    })

    output$table_potential_summary <- renderDT({
            not_b  <-  kh_tagged()  %>%  filter(Status  ==  "Chua  mua")  %>%  group_by(XHKH)  %>%
summarise(KH_Chua = n(), .groups = "drop")
      bought <- kh_tagged() %>% filter(Status == "Da mua") %>% group_by(XHKH) %>% summarise(KH_Da
= n(), .groups = "drop")
      df <- left_join(not_b, bought, by = "XHKH") %>% left_join(avg_spend(), by = "XHKH") %>%
          mutate(Ty_le = round(KH_Da / (KH_Da + KH_Chua) * 100, 1), DT_Pot = KH_Chua * Avg) %>%
arrange(desc(DT_Pot))
        datatable(df, colnames = c("Xep hang","KH chua mua","KH da mua","Chi tieu TB","Ty le CD
(%)","DT tiem nang"),
        rownames = FALSE, options = list(dom = "t", pageLength = 10,
                                                                                initComplete        =
JS("function(s,j){$(this.api().table().header()).css({'background-color':'#303983','color':'#fff',
'font-family':'Helvetica
Neue,sans-serif','font-weight':'600'});$(this.api().table().body()).css({'font-family':'Helvetica
Neue,sans-serif'});}"))) %>%
        formatCurrency(c("Avg","DT_Pot"), currency = "", interval = 3, mark = ".", digits = 0) %>%
        formatRound(c("KH_Chua","KH_Da"), digits = 0, interval = 3, mark = ".")
    })
  })
}
```

Source: phase7.R

```
phase7UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 7: Chien luoc kinh doanh thuc chien",
         style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
      p("Phan tich BCG Matrix, co hoi Up-sell/Cross-sell va tiem nang tang truong dua tren du lieu.",
           style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),

    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Ma tran BCG: Nhom san pham (Volume  Phi trung binh  Doanh thu)",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Kich thuoc bubble = tong GWP. Duong dut = trung binh. Goc phan tu: Star / Cash Cow / Dau hoi / Dog.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_bcg"), height = "480px"))
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Cross-sell: Nhom san pham  Xep hang khach hang",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Heatmap: vung dam = nhieu goi ban ra  co hoi cross-sell o vung nhat.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_cross_sell"), height = "390px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Ty le tai tuc theo nhom san pham",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Nhom nao giu chan khach hang tot nhat? Nhom nao can cai thien?",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_renewal_rate"), height = "390px"))
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Phi bao hiem trung binh theo xep hang khach hang",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Khach hang hang cao chi tra phi binh quan cao hon bao nhieu?",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_avg_premium_xhkh"), height = "390px")),
```

```
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("So goi trung binh moi khach hang theo xep hang",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Hang cao co cross-sell nhieu hon khong? (Goi TB/KH)",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_avg_packages_xhkh"), height = "390px"))
    ),

    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Ty le hop dong Moi vs Tai tuc theo nhom san pham",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Nhom nao chu yeu la HD moi? Nhom nao da co nen tai tuc?",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_new_vs_renew_group"), height = "390px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Doanh thu theo nhom san pham: Moi vs Tai tuc (Stacked)",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Phan dong gop GWP tu HD moi va tai tuc tai moi nhom SP.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_gwp_new_renew_stack"), height = "390px"))
    ),

    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Co hoi Up-sell: Khach hang phi thap nhung gia tri tai san cao",
                                   style  =  "color:#303983;  font-weight:700;  margin:0  0  2px  0;
font-family:'Helvetica Neue',sans-serif;"),
           tags$p("Xe gia tri > 500 trieu, phi BH < 3 trieu  dau hieu thieu bao ve, can tiep can
up-sell.",
                  style = "color:#6C7A89; font-size:12px; font-style:italic; margin-bottom:6px;"),
          DTOutput(ns("table_upsell_leads")))
    )
  )
}

phase7Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {
    get_data <- reactive({ req(all_data()); all_data()$master })
    pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")
       pal  <-  c("#4A6FA5",  "#2EC4B6",  "#7E57C2",  "#F06292",  "#FFCA28",  "#55bded",  "#66BB6A",
"#EF5350", "#95A5A6")
    ly <- function(p, bl = 10, bb = 10, bt = 10, br = 10) {
      p %>% layout(font = pf, margin = list(l = bl, r = br, t = bt, b = bb),
                          xaxis  =  list(automargin  =  TRUE),  yaxis  =  list(automargin  =  TRUE))  %>%
config(displayModeBar = FALSE)
    }
```

```
    output$plot_bcg <- renderPlotly({
      df_strat <- get_data() %>%
        filter(NHOMSANPHAM != "Chua xac dinh", !is.na(NHOMSANPHAM)) %>%
        group_by(NHOMSANPHAM) %>%
           summarise(Volume = n(), Revenue = sum(PHIBH, na.rm = TRUE), Avg_Premium = Revenue /
Volume, .groups = "drop")
      avg_vol <- mean(df_strat$Volume); avg_prem <- mean(df_strat$Avg_Premium)
      df_strat <- df_strat %>%
        mutate(Quadrant = case_when(
          Volume >= avg_vol & Avg_Premium >= avg_prem ~ "Ngoi sao (Star)",
          Volume >= avg_vol & Avg_Premium <  avg_prem ~ "Bo sua (Cash Cow)",
          Volume <  avg_vol & Avg_Premium >= avg_prem ~ "Dau hoi (?)",
          TRUE ~ "Cho (Dog)"))

      plot_ly(df_strat, x = ~Volume, y = ~Avg_Premium, size = ~Revenue, color = ~Quadrant,
               text = ~paste0(NHOMSANPHAM, "\nVolume: ", formatC(Volume, format = "d", big.mark =
"."),
                                                 "\nPhi TB: ", label_trieu(Avg_Premium), "\nGWP: ",
label_trieu(Revenue)),
              hoverinfo = "text", type = "scatter", mode = "markers",
              sizes = c(60, 250), # Tang kich thuoc bubble o day
              marker = list(opacity = 0.8, line = list(color = "#303983", width = 1.5)),
              colors = c("Ngoi sao (Star)" = "#2EC4B6", "Bo sua (Cash Cow)" = "#4A6FA5",
                         "Dau hoi (?)" = "#FFCA28", "Cho (Dog)" = "#F06292")) %>%
        add_annotations(x = df_strat$Volume, y = df_strat$Avg_Premium,
                        text = df_strat$NHOMSANPHAM, showarrow = FALSE,
                        font = list(size = 11, color = "#303983", family = "Helvetica Neue"),
                        yshift = 30) %>% # Tang khoang cach nhan ten san pham so voi bubble
        layout(
          font = pf,
          shapes = list(
            list(type = "line", x0 = avg_vol, x1 = avg_vol, y0 = 0, y1 = max(df_strat$Avg_Premium)
* 1.15,
                 line = list(color = "#303983", width = 1, dash = "dash")),
               list(type = "line", x0 = 0, x1 = max(df_strat$Volume) * 1.1, y0 = avg_prem, y1 =
avg_prem,
                 line = list(color = "#303983", width = 1, dash = "dash"))
          ),
          annotations = list(
            # Dung he toa do 'paper' de ep cac nhan phan loai ra 4 goc, tranh de len diem du lieu
            list(x = 0.98, y = 0.98, xref = "paper", yref = "paper", text = " Star",
                 showarrow = FALSE, font = list(size = 13, color = "#2EC4B6", family = "Helvetica
Neue"), xanchor = "right"),
            list(x = 0.98, y = 0.05, xref = "paper", yref = "paper", text = "Cash Cow",
                 showarrow = FALSE, font = list(size = 13, color = "#4A6FA5", family = "Helvetica
Neue"), xanchor = "right"),
            list(x = 0.02, y = 0.98, xref = "paper", yref = "paper", text = "? Dau hoi",
                 showarrow = FALSE, font = list(size = 13, color = "#FFCA28", family = "Helvetica
Neue"), xanchor = "left"),
            list(x = 0.02, y = 0.05, xref = "paper", yref = "paper", text = "Dog",
```

```
                  showarrow = FALSE, font = list(size = 13, color = "#F06292", family = "Helvetica
Neue"), xanchor = "left")
          ),
            xaxis = list(title = list(text = "So luong goi BH (Volume)", font = list(size = 11,
color = "#4A4A4A")),
                         gridcolor = "#EBEBEB", separatethousands = TRUE),
          yaxis = list(title = list(text = "Phi BH trung binh (VND)", font = list(size = 11, color
= "#4A4A4A")),
                         gridcolor = "#EBEBEB", tickformat = ",.0f"),
            legend = list(orientation = "h", x = 0.5, xanchor = "center", y = -0.22, yanchor =
"top", font = list(size = 10)), # Ha thap legend
          margin = list(l = 10, r = 10, t = 15, b = 80) # Tang le duoi tranh de legend vao title
        ) %>% config(displayModeBar = FALSE)
    })

    output$plot_cross_sell <- renderPlotly({
      df_cs <- get_data() %>% filter(NHOMSANPHAM != "Chua xac dinh") %>%
        group_by(NHOMSANPHAM, XHKH) %>% summarise(N = n(), .groups = "drop")
      p <- ggplot(df_cs, aes(x = XHKH, y = NHOMSANPHAM, fill = N)) +
        geom_tile(color = "white", linewidth = 0.6) +
        scale_fill_gradient(low = "#f0f2f7", high = "#303983") +
        theme_minimal(base_family = "sans") +
         theme(axis.text.x = element_text(angle = 25, hjust = 1, face = "bold", size = 9, color =
"#4A4A4A"),
              axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
                      panel.grid = element_blank(), axis.title = element_blank(), plot.margin =
margin(0,0,0,0))
      ggplotly(p, tooltip = c("x","y","fill")) %>% ly(bl = 10, bb = 45)
    })

    output$plot_renewal_rate <- renderPlotly({
      df_rr <- get_data() %>% filter(NHOMSANPHAM != "Chua xac dinh") %>%
        group_by(NHOMSANPHAM) %>%
         summarise(Total = n(), Pct = round(sum(LOAI_HD == "Tai tuc", na.rm = TRUE) / Total * 100,
1), .groups = "drop")
      p <- ggplot(df_rr, aes(x = reorder(NHOMSANPHAM, Pct), y = Pct)) +
        geom_col(fill = "#2EC4B6", width = 0.62) +
          geom_text(aes(label = paste0(Pct, "%")), hjust = -0.08, size = 3.3, color = "#303983",
fontface = "bold") +
        coord_flip() + scale_y_continuous(expand = expansion(mult = c(0, 0.18))) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.y = element_blank(), panel.grid.major.x = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), axis.text.y = element_text(face = "bold", size =
9, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "Ty le tai tuc (%)")
      ggplotly(p, tooltip = "y") %>% ly(bb = 25)
    })
```

```
    output$plot_avg_premium_xhkh <- renderPlotly({
       df_avg <- get_data() %>% group_by(XHKH) %>% summarise(Avg_Phi = mean(PHIBH, na.rm = TRUE),
.groups = "drop") %>%
        mutate(XHKH = factor(XHKH, levels = c("Dong", "Bac", "Vang", "Bach kim", "Kim cuong")))
      p <- ggplot(df_avg, aes(x = XHKH, y = Avg_Phi, fill = XHKH)) +
        geom_col(width = 0.6) +
        geom_text(aes(label = label_trieu(Avg_Phi)), vjust = -0.5, size = 3.3, color = "#303983",
fontface = "bold") +
             scale_fill_manual(values = c("Dong"="#95A5A6","Bac"="#BDC3C7","Vang"="#FFCA28","Bach
kim"="#7E57C2","Kim cuong"="#303983")) +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.18))) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.x = element_blank(), panel.grid.major.y = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), legend.position = "none",
              axis.text.x = element_text(face = "bold", size = 10, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "Phi BH trung binh (VND)")
      ggplotly(p, tooltip = "y") %>% ly(bb = 25)
    })

    output$plot_avg_packages_xhkh <- renderPlotly({
       df_pkg <- get_data() %>% group_by(XHKH, MA_KH) %>% summarise(N_goi = n(), .groups = "drop")
%>%
        group_by(XHKH) %>% summarise(Avg_goi = round(mean(N_goi), 2), .groups = "drop") %>%
        mutate(XHKH = factor(XHKH, levels = c("Dong", "Bac", "Vang", "Bach kim", "Kim cuong")))
      p <- ggplot(df_pkg, aes(x = XHKH, y = Avg_goi, fill = XHKH)) +
        geom_col(width = 0.6) +
         geom_text(aes(label = Avg_goi), vjust = -0.5, size = 3.5, color = "#303983", fontface =
"bold") +
             scale_fill_manual(values = c("Dong"="#95A5A6","Bac"="#BDC3C7","Vang"="#FFCA28","Bach
kim"="#7E57C2","Kim cuong"="#303983")) +
        scale_y_continuous(expand = expansion(mult = c(0, 0.18))) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.x = element_blank(), panel.grid.major.y = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), legend.position = "none",
              axis.text.x = element_text(face = "bold", size = 10, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "So goi TB / khach hang")
      ggplotly(p, tooltip = "y") %>% ly(bb = 25)
    })

    output$plot_new_vs_renew_group <- renderPlotly({
      df_nr <- get_data() %>% filter(NHOMSANPHAM != "Chua xac dinh") %>%
        group_by(NHOMSANPHAM, LOAI_HD) %>% summarise(N = n(), .groups = "drop")
      p <- ggplot(df_nr, aes(x = NHOMSANPHAM, y = N, fill = LOAI_HD)) +
```

```
        geom_col(position = "fill", width = 0.62) +
          scale_y_continuous(labels = scales::percent_format(), expand = expansion(mult = c(0,
0.02))) +
        scale_fill_manual(values = c("Moi" = "#4A6FA5", "Tai tuc" = "#FFCA28")) +
        theme_minimal(base_family = "sans") +
         theme(axis.text.x = element_text(angle = 25, hjust = 1, face = "bold", size = 9, color =
"#4A4A4A"),
                  panel.grid.major.x = element_blank(), panel.grid.major.y = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), legend.title = element_blank(),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "Ty le (%)")
      ggplotly(p) %>%
        layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 55),
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom")) %>% config(displayModeBar = FALSE)
    })

    output$plot_gwp_new_renew_stack <- renderPlotly({
      df_gwp <- get_data() %>% filter(NHOMSANPHAM != "Chua xac dinh") %>%
          group_by(NHOMSANPHAM, LOAI_HD) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups =
"drop")
      p <- ggplot(df_gwp, aes(x = reorder(NHOMSANPHAM, -GWP), y = GWP, fill = LOAI_HD)) +
        geom_col(width = 0.62) +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.05))) +
        scale_fill_manual(values = c("Moi" = "#4A6FA5", "Tai tuc" = "#FFCA28")) +
        theme_minimal(base_family = "sans") +
         theme(axis.text.x = element_text(angle = 25, hjust = 1, face = "bold", size = 9, color =
"#4A4A4A"),
                  panel.grid.major.x = element_blank(), panel.grid.major.y = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), legend.title = element_blank(),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "GWP (VND)")
      ggplotly(p) %>%
        layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 55),
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom")) %>% config(displayModeBar = FALSE)
    })

    output$table_upsell_leads <- renderDT({
      upsell <- get_data() %>%
        filter(!is.na(GIA_TIEN_XE), GIA_TIEN_XE > 500000000, PHIBH < 3000000) %>%
         select(MA_KH, MA_HD, TEN_DONG_XE, TEN_HANG_XE, GIA_TIEN_XE, PHIBH, TENGOISANPHAM, TEN_CN)
%>%
        arrange(desc(GIA_TIEN_XE)) %>% head(100)
      datatable(upsell,
         colnames = c("Ma KH","Ma HD","Dong xe","Hang","Gia tri xe","Phi BH","Goi dang dung","Chi
```

```
nhanh"),
        rownames = FALSE, options = list(pageLength = 8, scrollX = TRUE,
                                                                             initComplete        =
JS("function(s,j){$(this.api().table().header()).css({'background-color':'#303983','color':'#fff',
'font-family':'Helvetica
Neue,sans-serif','font-weight':'600'});$(this.api().table().body()).css({'font-family':'Helvetica
Neue,sans-serif'});}"))) %>%
        formatCurrency(c("GIA_TIEN_XE","PHIBH"), currency = "", interval = 3, mark = ".", digits =
0)
    })
  })
}
```

Source: phase8.R

```
phase8UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 8: Quan tri KPI chuyen sau",
         style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    p("Phan tich chi tiet muc do hoan thanh chi tieu theo tung don vi, kenh ban va san pham.",
      style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),

    # --- KPI Cards ---
    fluidRow(
      valueBoxOutput(ns("vbox_kpi_overall"), width = 4),
      valueBoxOutput(ns("vbox_kpi_sl"), width = 4),
      valueBoxOutput(ns("vbox_kpi_gap"), width = 4)
    ),

    # --- Row 1: KPI monthly ---
    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Hoan thanh KPI doanh thu theo thang (Thuc te vs Muc tieu)",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Bar steel blue = thuc te, bar hong nhat = muc tieu KPI.",
                 style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_kpi_monthly"), height = "410px"))
    ),

    # --- Row 2: Kenh ban + San pham ---
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("KPI theo kenh ban (Thuc te vs Muc tieu)",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("So sanh doanh thu thuc te va chi tieu theo tung kenh.",
                 style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_kpi_channel"), height = "390px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("KPI theo goi san pham (Top 10)",
                  style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Top 10 goi san pham co doanh thu thuc te cao nhat.",
                 style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_kpi_product"), height = "390px"))
    ),

    # --- Row 3 MOI: % hoan thanh theo kenh (bullet-style) + Gap phan tich ---
```

```
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Ty le hoan thanh KPI doanh thu theo kenh (%)",
                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Duong dut net do = moc 100%. Kenh nao vuot, kenh nao thieu?",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_kpi_pct_channel"), height = "390px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Chenh lech doanh thu (Actual  Target) theo kenh",
                                   style = "color:#303983; font-weight:700; margin:0 0 2px 0;
font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Xanh = vuot KPI, hong = chua dat. Gia tri tuyet doi.",
                  style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_kpi_gap_channel"), height = "390px"))
    ),

    # --- Row 4: Bang chi tiet ---
    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Chi tiet KPI theo kenh  thang",
                                   style = "color:#303983; font-weight:700; margin:0 0 6px 0;
font-family:'Helvetica Neue',sans-serif;"),
          DTOutput(ns("table_kpi_detail")))
    )
  )
}

phase8Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {

    get_data <- reactive({ req(all_data()); all_data()$master })
    get_kpi  <- reactive({ req(all_data()); all_data()$kpi })

    pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")

    ly <- function(p, bl = 10, bb = 10, bt = 10, br = 10) {
      p %>% layout(font = pf, margin = list(l = bl, r = br, t = bt, b = bb),
                   xaxis = list(automargin = TRUE), yaxis = list(automargin = TRUE)
      ) %>% config(displayModeBar = FALSE)
    }


    output$vbox_kpi_overall <- renderValueBox({
      total_act <- sum(get_data()$PHIBH, na.rm = TRUE)
      total_target <- sum(get_kpi()$DOANH_THU, na.rm = TRUE)
      pct <- ifelse(total_target > 0, round(total_act / total_target * 100, 1), 0)
      valueBox(paste0(pct, "%"), "Ty le hoan thanh doanh thu", icon = icon("check-double"),
               color = ifelse(pct >= 100, "green", ifelse(pct >= 80, "yellow", "red")))
    })
```

```
    output$vbox_kpi_sl <- renderValueBox({
      total_target_sl <- sum(get_kpi()$SLHD, na.rm = TRUE)
      pct <- ifelse(total_target_sl > 0, round(nrow(get_data()) / total_target_sl * 100, 1), 0)
        valueBox(paste0(pct, "%"), "Ty le hoan thanh so luong", icon = icon("file-alt"), color =
"blue")
    })

    output$vbox_kpi_gap <- renderValueBox({
      gap <- sum(get_kpi()$DOANH_THU, na.rm = TRUE) - sum(get_data()$PHIBH, na.rm = TRUE)
      valueBox(format_vnd(abs(gap)),
               ifelse(gap > 0, "Con thieu so voi KPI", "Vuot KPI"),
               icon = icon(ifelse(gap > 0, "arrow-down", "arrow-up")),
               color = ifelse(gap > 0, "red", "green"))
    })


    output$plot_kpi_monthly <- renderPlotly({
      act_sum <- get_data() %>% mutate(YM = format(NGAY_KY_HD, "%Y%m")) %>%
        group_by(YM) %>% summarise(Actual = sum(PHIBH, na.rm = TRUE), .groups = "drop")
      kpi_sum <- get_kpi() %>% group_by(YM = as.character(THANGNAM)) %>%
        summarise(Target = sum(DOANH_THU, na.rm = TRUE), .groups = "drop")

      df_combo <- full_join(act_sum, kpi_sum, by = "YM") %>% arrange(YM) %>%
        mutate(Label = paste0(substr(YM, 5, 6), "/", substr(YM, 1, 4)))

      plot_ly(df_combo, x = ~Label) %>%
        add_bars(y = ~Actual, name = "Thuc te", marker = list(color = "#4A6FA5")) %>%
         add_bars(y = ~Target, name = "Muc tieu (KPI)", marker = list(color = "#F06292", opacity =
0.6)) %>%
        layout(
          font = pf,
          yaxis = list(title = list(text = "Doanh thu (VND)", font = list(size = 11)),
                       tickformat = ",.0f", gridcolor = "#EBEBEB"),
          xaxis = list(title = "", tickangle = -45, gridcolor = "#EBEBEB"),
          barmode = "group", bargap = 0.25,
            legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom"),
          margin = list(l = 10, r = 10, t = 40, b = 45)
        ) %>% config(displayModeBar = FALSE)
    })


    output$plot_kpi_channel <- renderPlotly({
      df_act <- get_data() %>% filter(TENKENHBAN != "Chua xac dinh") %>%
         group_by(MA_KENHBAN, TENKENHBAN) %>% summarise(Actual = sum(PHIBH, na.rm = TRUE), .groups
= "drop")
      df_kpi <- get_kpi() %>% group_by(MA_KENHBAN) %>%
        summarise(Target = sum(DOANH_THU, na.rm = TRUE), .groups = "drop")
```

```
      df_long <- left_join(df_act, df_kpi, by = "MA_KENHBAN") %>%
        select(TENKENHBAN, Actual, Target) %>%
        pivot_longer(cols = c(Actual, Target), names_to = "Type", values_to = "Value")

      p <- ggplot(df_long, aes(x = reorder(TENKENHBAN, Value), y = Value, fill = Type)) +
        geom_col(position = "dodge", width = 0.7) + coord_flip() +
        scale_fill_manual(values = c("Actual" = "#4A6FA5", "Target" = "#F06292"),
                          labels = c("Thuc te", "Muc tieu")) +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
        theme_minimal(base_family = "sans") +
        theme(
          panel.grid.major.y = element_blank(),
            panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
          panel.grid.minor = element_blank(),
          axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
          axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
          legend.title = element_blank(),
          plot.margin = margin(0, 0, 0, 0)
        ) + labs(x = "", y = "Doanh thu (VND)")

      ggplotly(p) %>%
        layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 20),
               xaxis = list(automargin = TRUE), yaxis = list(automargin = TRUE),
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom")
        ) %>% config(displayModeBar = FALSE)
    })


    output$plot_kpi_product <- renderPlotly({
      df_prod <- get_data() %>%
        filter(TENGOISANPHAM != "Chua xac dinh") %>%
        group_by(TENGOISANPHAM) %>% summarise(Actual = sum(PHIBH, na.rm = TRUE), .groups = "drop")
%>%
        arrange(desc(Actual)) %>% head(10)

      p <- ggplot(df_prod, aes(x = reorder(TENGOISANPHAM, Actual), y = Actual)) +
        geom_col(fill = "#4A6FA5", width = 0.68) + coord_flip() +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
        theme_minimal(base_family = "sans") +
        theme(
          panel.grid.major.y = element_blank(),
            panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
          panel.grid.minor = element_blank(),
          axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
          axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
          plot.margin = margin(0, 0, 0, 0)
        ) + labs(x = "", y = "Doanh thu thuc te (VND)")
```

```
      ggplotly(p) %>% ly(bb = 25)
    })

    output$plot_kpi_pct_channel <- renderPlotly({
      df_act <- get_data() %>% filter(TENKENHBAN != "Chua xac dinh") %>%
         group_by(MA_KENHBAN, TENKENHBAN) %>% summarise(Actual = sum(PHIBH, na.rm = TRUE), .groups
= "drop")
      df_kpi <- get_kpi() %>% group_by(MA_KENHBAN) %>%
        summarise(Target = sum(DOANH_THU, na.rm = TRUE), .groups = "drop")

      df_pct <- left_join(df_act, df_kpi, by = "MA_KENHBAN") %>%
        mutate(Pct = ifelse(!is.na(Target) & Target > 0, round(Actual / Target * 100, 1), 0)) %>%
        mutate(Color = ifelse(Pct >= 100, "#2EC4B6", "#F06292"))

      p <- ggplot(df_pct, aes(x = reorder(TENKENHBAN, Pct), y = Pct, fill = Pct >= 100)) +
        geom_col(width = 0.62) +
        geom_hline(yintercept = 100, linetype = "dashed", color = "#EF5350", linewidth = 0.7) +
          geom_text(aes(label = paste0(Pct, "%")), hjust = -0.08, size = 3.3, color = "#303983",
fontface = "bold") +
        coord_flip() +
        scale_fill_manual(values = c("TRUE" = "#2EC4B6", "FALSE" = "#F06292"), guide = "none") +
        scale_y_continuous(expand = expansion(mult = c(0, 0.2))) +
        theme_minimal(base_family = "sans") +
        theme(
          panel.grid.major.y = element_blank(),
            panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
          panel.grid.minor = element_blank(),
          axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
          axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
          plot.margin = margin(0, 0, 0, 0)
        ) + labs(x = "", y = "% Hoan thanh")

      ggplotly(p, tooltip = "y") %>% ly(bb = 25)
    })


    output$plot_kpi_gap_channel <- renderPlotly({
      df_act <- get_data() %>% filter(TENKENHBAN != "Chua xac dinh") %>%
         group_by(MA_KENHBAN, TENKENHBAN) %>% summarise(Actual = sum(PHIBH, na.rm = TRUE), .groups
= "drop")
      df_kpi <- get_kpi() %>% group_by(MA_KENHBAN) %>%
        summarise(Target = sum(DOANH_THU, na.rm = TRUE), .groups = "drop")

      df_gap <- left_join(df_act, df_kpi, by = "MA_KENHBAN") %>%
        mutate(Gap = Actual - ifelse(is.na(Target), 0, Target),
               Direction = ifelse(Gap >= 0, "Vuot", "Thieu"))

      p <- ggplot(df_gap, aes(x = reorder(TENKENHBAN, Gap), y = Gap, fill = Direction)) +
```

```
        geom_col(width = 0.62) +
        geom_hline(yintercept = 0, color = "#303983", linewidth = 0.5) +
        coord_flip() +
        scale_fill_manual(values = c("Vuot" = "#2EC4B6", "Thieu" = "#F06292"), guide = "none") +
        scale_y_continuous(labels = label_trieu) +
        theme_minimal(base_family = "sans") +
        theme(
          panel.grid.major.y = element_blank(),
            panel.grid.major.x = element_line(color = "#EBEBEB", linetype = "dashed", linewidth =
0.35),
          panel.grid.minor = element_blank(),
          axis.text.y = element_text(face = "bold", size = 9, color = "#4A4A4A"),
          axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"),
          plot.margin = margin(0, 0, 0, 0)
        ) + labs(x = "", y = "Chenh lech (VND)")

      ggplotly(p) %>% ly(bb = 25)
    })


    output$table_kpi_detail <- renderDT({
      df_act <- get_data() %>% mutate(YM = format(NGAY_KY_HD, "%Y%m")) %>%
        group_by(YM, MA_KENHBAN, TENKENHBAN) %>%
        summarise(Actual_DT = sum(PHIBH, na.rm = TRUE), Actual_SL = n(), .groups = "drop")
      df_kpi <- get_kpi() %>% group_by(YM = as.character(THANGNAM), MA_KENHBAN) %>%
         summarise(Target_DT = sum(DOANH_THU, na.rm = TRUE), Target_SL = sum(SLHD, na.rm = TRUE),
.groups = "drop")

      df_detail <- full_join(df_act, df_kpi, by = c("YM", "MA_KENHBAN")) %>%
        mutate(
            Pct_DT = ifelse(!is.na(Target_DT) & Target_DT > 0, round(Actual_DT / Target_DT * 100,
1), NA),
            Pct_SL = ifelse(!is.na(Target_SL) & Target_SL > 0, round(Actual_SL / Target_SL * 100,
1), NA)
        ) %>%
        arrange(YM, MA_KENHBAN) %>%
        select(YM, TENKENHBAN, Actual_DT, Target_DT, Pct_DT, Actual_SL, Target_SL, Pct_SL)

      datatable(df_detail,
        colnames = c("Thang", "Kenh ban", "DT thuc te", "DT muc tieu", "% Hoan thanh DT",
                     "SL thuc te", "SL muc tieu", "% Hoan thanh SL"),
        filter = "top", rownames = FALSE,
        options = list(pageLength = 10, scrollX = TRUE,
          initComplete = JS(
            "function(settings, json) {",
            "  $(this.api().table().header()).css({",
            "    'background-color':'#303983','color':'#fff',",
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif','font-weight':'600'",
            "  });",
            "  $(this.api().table().body()).css({",
```

```
            "    'font-family':'Helvetica Neue,Helvetica,Arial,sans-serif'",
            "  });",
            "}"
          )
        )
      ) %>%
           formatCurrency(c("Actual_DT", "Target_DT"), currency = "", interval = 3, mark = ".",
digits = 0) %>%
        formatStyle("Pct_DT",
          backgroundColor = styleInterval(c(80, 100), c("#FADBD8", "#FCF3CF", "#D5F5E3")),
          fontWeight = "bold")
    })
  })
}
```

Source: phase9.R

```
phase9UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 9: Quan tri khach hang tiem nang va tai tuc",
            style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    p("Canh bao som hop dong sap het han, goi y tep khach hang tiem nang va phan tich pheu chuyen doi.",
            style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
    fluidRow(
      valueBoxOutput(ns("vbox_expiring_30d"), width = 3), valueBoxOutput(ns("vbox_expiring_60d"), width = 3),
      valueBoxOutput(ns("vbox_high_value_leads"), width = 3), valueBoxOutput(ns("vbox_renewal_rate"), width = 3)
    ),
    fluidRow(
      box(title = NULL, width = 12, status = "primary", solidHeader = FALSE,
          tags$h4("Goi BH het han trong 60 ngay toi", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Danh sach khach hang can lien he gap de duy tri ty le tai tuc.", style = "color:#EF5350; font-size:12px; font-weight:bold; font-style:italic; margin-bottom:6px;"),
          DTOutput(ns("table_renewal_leads")))
    ),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Tep khach hang tiem nang (Hang cao)", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Kim cuong/Bach kim  chua da dang goi san pham.", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          DTOutput(ns("table_high_value_leads"))),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Pheu chuyen doi du kien", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Tu tong KH  hang cao  sap het han  da tai tuc.", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_lead_funnel"), height = "380px"))
    ),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Phan bo ngay het han theo thang (12 thang toi)", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
          tags$p("Bar = so goi het han, line = xu huong.", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_expiry_timeline"), height = "380px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
          tags$h4("Doanh thu tai tuc tiem nang theo nhom san pham", style = "color:#303983;
```

```
font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
              tags$p("GWP tiem nang neu tai tuc thanh cong toan bo goi sap het han.", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_renewal_value"), height = "380px"))
    ),
    fluidRow(
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
                  tags$h4("Ty le tai tuc theo chi nhanh (Top 15)", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
                tags$p("Chi nhanh nao giu chan khach hang tot nhat?", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_renewal_by_cn"), height = "380px")),
      box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
            tags$h4("Phan bo thoi han con lai cua hop dong (Histogram)", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
            tags$p("Duong dut do = moc 30, 60, 90 ngay.", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_ttl_histogram"), height = "380px"))
    )
  )
}

phase9Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {
    get_data <- reactive({ req(all_data()); all_data()$master })
    today_date <- as.Date("2026-04-16")
    pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")
    pal <- c("#303983", "#4A6FA5", "#2EC4B6", "#F06292", "#FFCA28", "#55bded")
    ly <- function(p, bl = 10, bb = 10, bt = 10, br = 10) {
      p %>% layout(font = pf, margin = list(l = bl, r = br, t = bt, b = bb),
                          xaxis = list(automargin = TRUE), yaxis = list(automargin = TRUE)) %>% config(displayModeBar = FALSE)
    }

    output$vbox_expiring_30d <- renderValueBox({
      n <- get_data() %>% filter(NGAY_HET_HAN >= today_date, NGAY_HET_HAN <= today_date + 30) %>% nrow()
          valueBox(formatC(n, format = "d", big.mark = "."), "Het han < 30 ngay", icon = icon("clock"), color = "red")
    })
    output$vbox_expiring_60d <- renderValueBox({
      n <- get_data() %>% filter(NGAY_HET_HAN >= today_date, NGAY_HET_HAN <= today_date + 60) %>% nrow()
          valueBox(formatC(n, format = "d", big.mark = "."), "Het han < 60 ngay", icon = icon("hourglass-half"), color = "yellow")
    })
    output$vbox_high_value_leads <- renderValueBox({
        n <- get_data() %>% filter(XHKH %in% c("Kim cuong", "Bach kim")) %>% pull(MA_KH) %>% n_distinct()
      valueBox(formatC(n, format = "d", big.mark = "."), "KH hang cao", icon = icon("gem"), color
```

```
= "purple")
    })
    output$vbox_renewal_rate <- renderValueBox({
      df <- get_data(); pct <- round(sum(df$LOAI_HD == "Tai tuc", na.rm = TRUE) / nrow(df) * 100,
1)
      valueBox(paste0(pct, "%"), "Ty le tai tuc", icon = icon("sync"), color = "green")
    })

    output$table_renewal_leads <- renderDT({
      renewal <- get_data() %>%
        filter(NGAY_HET_HAN >= today_date, NGAY_HET_HAN <= (today_date + 60)) %>%
        mutate(So_ngay = as.numeric(NGAY_HET_HAN - today_date),
                Muc_do = case_when(So_ngay <= 7 ~ "Khan cap", So_ngay <= 30 ~ "Can xu ly", TRUE ~
"Theo doi")) %>%
        select(MA_KH, MA_HD, TENGOISANPHAM, TEN_CN, PHIBH, NGAY_HET_HAN, So_ngay, Muc_do, SDT_NV)
%>%
        arrange(So_ngay) %>% head(200)
        datatable(renewal, colnames = c("Ma KH","Ma HD","Goi SP","Chi nhanh","Phi BH","Ngay het
han","So ngay con","Muc do","SDT NV"),
        rownames = FALSE, options = list(pageLength = 10, scrollX = TRUE,
                                                                    initComplete =
JS("function(s,j){$(this.api().table().header()).css({'background-color':'#303983','color':'#fff',
'font-family':'Helvetica
Neue,sans-serif','font-weight':'600'});$(this.api().table().body()).css({'font-family':'Helvetica
Neue,sans-serif'});}"))) %>%
        formatCurrency("PHIBH", currency = "", interval = 3, mark = ".", digits = 0) %>%
         formatStyle("Muc_do", backgroundColor = styleEqual(c("Khan cap","Can xu ly","Theo doi"),
c("#EF5350","#FFCA28","#55bded")), color = "white", fontWeight = "bold")
    })

    output$table_high_value_leads <- renderDT({
      high_val <- get_data() %>% filter(XHKH %in% c("Kim cuong", "Bach kim")) %>%
        group_by(MA_KH, XHKH, TEN_CN) %>%
                summarise(So_goi = n(), Tong_Phi = sum(PHIBH, na.rm = TRUE), SP_da_co =
paste(unique(NHOMSANPHAM), collapse = ", "), .groups = "drop") %>%
        arrange(desc(Tong_Phi)) %>% head(50)
       datatable(high_val, colnames = c("Ma KH","Xep hang","Chi nhanh","So goi","Tong phi","SP da
co"),
        rownames = FALSE, options = list(pageLength = 8, scrollX = TRUE,
                                                                    initComplete =
JS("function(s,j){$(this.api().table().header()).css({'background-color':'#303983','color':'#fff',
'font-family':'Helvetica
Neue,sans-serif','font-weight':'600'});$(this.api().table().body()).css({'font-family':'Helvetica
Neue,sans-serif'});}"))) %>%
        formatCurrency("Tong_Phi", currency = "", interval = 3, mark = ".", digits = 0)
    })

    output$plot_lead_funnel <- renderPlotly({
      df <- get_data()
      plot_ly(type = "funnel",
```

```
              y = c("Tong KH unique", "KH hang Vang+", "KH sap het han (90d)", "KH da tai tuc"),
              x = c(n_distinct(df$MA_KH),
                     df %>% filter(XHKH %in% c("Kim cuong","Bach kim","Vang")) %>% pull(MA_KH) %>%
n_distinct(),
                    df %>% filter(NGAY_HET_HAN >= today_date, NGAY_HET_HAN <= today_date + 90) %>%
pull(MA_KH) %>% n_distinct(),
                    df %>% filter(LOAI_HD == "Tai tuc") %>% pull(MA_KH) %>% n_distinct()),
              textinfo = "value+percent initial", marker = list(color = pal[1:4])) %>%
               layout(font = pf, margin = list(l = 140, r = 20, t = 20, b = 20)) %>%
config(displayModeBar = FALSE)
    })

    output$plot_expiry_timeline <- renderPlotly({
        df_exp <- get_data() %>% filter(NGAY_HET_HAN >= today_date, NGAY_HET_HAN <= today_date +
365) %>%
        mutate(Thang_HH = floor_date(NGAY_HET_HAN, "month")) %>%
        group_by(Thang_HH) %>% summarise(N = n(), .groups = "drop")
      plot_ly(df_exp, x = ~Thang_HH) %>%
        add_bars(y = ~N, name = "So goi", marker = list(color = "#55bded", opacity = 0.6)) %>%
          add_lines(y = ~N, name = "Xu huong", line = list(color = "#303983", width = 2, shape =
"spline", smoothing = 1.3)) %>%
        layout(font = pf, margin = list(l = 10, r = 10, t = 40, b = 15),
                xaxis = list(automargin = TRUE, gridcolor = "#EBEBEB", tickformat = "%b %Y", dtick
= "M2"),
                  yaxis = list(automargin = TRUE, gridcolor = "#EBEBEB", title = list(text = "So
goi", font = list(size = 11)), separatethousands = TRUE),
                legend = list(orientation = "h", x = 0.5, xanchor = "center", y = 1.12, yanchor =
"bottom"), bargap = 0.3) %>% config(displayModeBar = FALSE)
    })

    output$plot_renewal_value <- renderPlotly({
      df_rv <- get_data() %>% filter(NGAY_HET_HAN >= today_date, NGAY_HET_HAN <= today_date + 365,
NHOMSANPHAM != "Chua xac dinh") %>%
        group_by(NHOMSANPHAM) %>% summarise(GWP = sum(PHIBH, na.rm = TRUE), .groups = "drop")
      p <- ggplot(df_rv, aes(x = reorder(NHOMSANPHAM, GWP), y = GWP)) +
        geom_col(fill = "#2EC4B6", width = 0.62) + coord_flip() +
        scale_y_continuous(labels = label_trieu, expand = expansion(mult = c(0, 0.08))) +
        theme_minimal(base_family = "sans") +
          theme(panel.grid.major.y = element_blank(), panel.grid.major.x = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), axis.text.y = element_text(face = "bold", size =
9, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "GWP tiem nang (VND)")
      ggplotly(p) %>% ly(bb = 25)
    })

    output$plot_renewal_by_cn <- renderPlotly({
        df_rr <- get_data() %>% filter(TEN_CN != "Chua xac dinh", !grepl("Hoi so|TCT", TEN_CN,
```

```
ignore.case = TRUE)) %>%
         group_by(TEN_CN) %>% summarise(Total = n(), Pct = round(sum(LOAI_HD == "Tai tuc", na.rm =
TRUE) / Total * 100, 1), .groups = "drop") %>%
        top_n(15, Total)
      p <- ggplot(df_rr, aes(x = reorder(TEN_CN, Pct), y = Pct)) +
        geom_col(fill = "#7E57C2", width = 0.62) +
          geom_text(aes(label = paste0(Pct, "%")), hjust = -0.08, size = 3.3, color = "#303983",
fontface = "bold") +
        coord_flip() + scale_y_continuous(expand = expansion(mult = c(0, 0.2))) +
        theme_minimal(base_family = "sans") +
           theme(panel.grid.major.y = element_blank(), panel.grid.major.x = element_line(color =
"#EBEBEB", linetype = "dashed", linewidth = 0.35),
              panel.grid.minor = element_blank(), axis.text.y = element_text(face = "bold", size =
9, color = "#4A4A4A"),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "", y = "Ty le tai tuc (%)")
      ggplotly(p, tooltip = "y") %>% ly(bb = 25)
    })

    output$plot_ttl_histogram <- renderPlotly({
      df_ttl <- get_data() %>% filter(NGAY_HET_HAN >= today_date) %>%
        mutate(TTL = as.numeric(NGAY_HET_HAN - today_date)) %>% filter(TTL <= 365)
      p <- ggplot(df_ttl, aes(x = TTL)) +
         geom_histogram(binwidth = 15, fill = "#4A6FA5", alpha = 0.8, color = "white", linewidth =
0.2) +
         geom_vline(xintercept = c(30, 60, 90), linetype = "dashed", color = "#EF5350", linewidth =
0.6) +
           annotate("text", x = 30, y = Inf, label = "30d", vjust = 2, hjust = -0.2, color =
"#EF5350", size = 3, fontface = "bold") +
           annotate("text", x = 60, y = Inf, label = "60d", vjust = 2, hjust = -0.2, color =
"#EF5350", size = 3, fontface = "bold") +
           annotate("text", x = 90, y = Inf, label = "90d", vjust = 2, hjust = -0.2, color =
"#EF5350", size = 3, fontface = "bold") +
         scale_y_continuous(expand = expansion(mult = c(0, 0.08)), labels = function(x) formatC(x,
format = "d", big.mark = ".")) +
        theme_minimal(base_family = "sans") +
         theme(panel.grid.major.y = element_line(color = "#EBEBEB", linetype = "dashed", linewidth
= 0.35),
              panel.grid.major.x = element_blank(), panel.grid.minor = element_blank(),
               axis.title = element_text(face = "bold", size = 10, color = "#4A4A4A"), plot.margin
= margin(0,0,0,0)) +
        labs(x = "So ngay con lai", y = "So goi BH")
      ggplotly(p) %>% ly(bb = 25)
    })
  })
}
```

Source: phase10.R

```
phase10UI <- function(id) {
  ns <- NS(id)
  tabItem(tabName = id,
    h2("Phan 10: Tong ket du an va De xuat dieu hanh",
         style = "color:#303983; font-weight:700; margin-bottom:4px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),
     p("Tom luoc cac phat hien chinh, chi so cot loi va dinh huong chien luoc cho giai doan tiep theo.",
                   style = "color:#6C7A89; margin-bottom:18px; font-family:'Helvetica Neue',Helvetica,Arial,sans-serif;"),

    fluidRow(
      valueBoxOutput(ns("vb_total_gwp"), width = 3), valueBoxOutput(ns("vb_total_hd"), width = 3),
      valueBoxOutput(ns("vb_total_kh"), width = 3), valueBoxOutput(ns("vb_renewal"), width = 3)
    ),

    fluidRow(
      box(title = NULL, width = 8, status = "primary", solidHeader = FALSE,
           tags$h4("Hieu suat tong the du an", style = "color:#303983; font-weight:700; margin:0 0 2px 0; font-family:'Helvetica Neue',sans-serif;"),
               tags$p("Danh gia muc do hoan thanh cac chi so loi so voi ky vong.", style = "color:#6C7A89; font-size:12px; margin-bottom:6px;"),
          plotlyOutput(ns("plot_radar_summary"), height = "400px")),
      box(title = NULL, width = 4, status = "primary", solidHeader = FALSE,
            tags$h4("Chi so cot loi", style = "color:#303983; font-weight:700; margin:0 0 8px 0; font-family:'Helvetica Neue',sans-serif;"),
          uiOutput(ns("ui_core_metrics")))
    ),

    tags$h3("Ket luan chinh",
               style = "color:#303983; font-weight:700; font-size:22px; margin:25px 0 15px 0; font-family:'Helvetica Neue',sans-serif; border-bottom:2px solid #EBEBEB; padding-bottom:8px;"),
    fluidRow(
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE, uiOutput(ns("ui_conclusion_sales"))),
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE, uiOutput(ns("ui_conclusion_customer")))
    ),
    fluidRow(
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE, uiOutput(ns("ui_conclusion_product"))),
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE, uiOutput(ns("ui_conclusion_challenge")))
    ),

    tags$h3("Khuyen nghi chien luoc",
               style = "color:#303983; font-weight:700; font-size:22px; margin:25px 0 15px 0;
```

```
font-family:'Helvetica Neue',sans-serif; border-bottom:2px solid #EBEBEB; padding-bottom:8px;"),
    fluidRow(
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
uiOutput(ns("ui_recom_retention"))),
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
uiOutput(ns("ui_recom_product")))
    ),
    fluidRow(
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
uiOutput(ns("ui_recom_channel"))),
              box(title = NULL, width = 6, status = "primary", solidHeader = FALSE,
uiOutput(ns("ui_recom_tech")))
    ),

    fluidRow(
      column(12, tags$div(style = "text-align:center; padding:40px 0 20px 0;",
                tags$h2("Cam on!", style = "color:#303983; font-weight:700; font-size:32px;
margin-bottom:8px; font-family:'Helvetica Neue',sans-serif;"),
        tags$p("Nhom: ", tags$strong("10tr_devided_by_5"), style = "font-size:15px; color:#4A4A4A;
font-family:'Helvetica Neue',sans-serif;"),
        tags$p("tam thoi de trong mail", style = "color:#4A6FA5; font-weight:bold; font-size:15px;
font-family:'Helvetica Neue',sans-serif;")
      ))
    )
  )
}

phase10Server <- function(id, all_data) {
  moduleServer(id, function(input, output, session) {
    get_data <- reactive({ req(all_data()); all_data()$master })
    get_kpi  <- reactive({ req(all_data()); all_data()$kpi })
    pf <- list(family = "'Helvetica Neue', Helvetica, Arial, sans-serif", color = "#4A4A4A")
          css_h <- "color:#303983; font-weight:700; font-size:16px; margin:0 0 10px 0;
font-family:'Helvetica Neue',sans-serif;"
          css_b <- "font-family:'Helvetica Neue',sans-serif; color:#4A4A4A; font-size:13px;
line-height:1.9;"
    css_s <- "color:#303983;"

    output$vb_total_gwp <- renderValueBox({
      valueBox(format_vnd(sum(get_data()$PHIBH, na.rm = TRUE)), "Tong GWP", color = "blue")
    })
    output$vb_total_hd <- renderValueBox({
        valueBox(formatC(n_distinct(get_data()$MA_HD), format = "d", big.mark = "."), "Hop dong",
color = "blue")
    })
    output$vb_total_kh <- renderValueBox({
       valueBox(formatC(n_distinct(get_data()$MA_KH), format = "d", big.mark = "."), "Khach hang",
color = "teal")
    })
    output$vb_renewal <- renderValueBox({
```

```
      df <- get_data(); pct <- round(sum(df$LOAI_HD == "Tai tuc", na.rm = TRUE) / nrow(df) * 100, 1)
      valueBox(paste0(pct, "%"), "Ty le tai tuc", color = "green")
    })

    output$plot_radar_summary <- renderPlotly({
      df <- get_data(); kpi <- get_kpi()
      kpi_pct <- min(round(sum(df$PHIBH, na.rm = TRUE) / sum(kpi$DOANH_THU, na.rm = TRUE) * 100, 0), 100)
      renewal_pct <- round(sum(df$LOAI_HD == "Tai tuc") / nrow(df) * 100, 0)
      n_cn <- n_distinct(df$TEN_CN[df$TEN_CN != "Chua xac dinh"])
      coverage <- min(round(n_cn / 22 * 100, 0), 100)
      avg_goi <- round(nrow(df) / n_distinct(df$MA_KH), 2)
      crosssell <- min(round(avg_goi / 3 * 100, 0), 100)
      scores <- c(kpi_pct, renewal_pct, coverage, crosssell, 85)
      labels <- c("Hoan thanh KPI", "Ty le tai tuc", "Phu song CN", "Cross-sell index", "Chat luong DL")
      plot_ly(type = "scatterpolar", r = c(scores, scores[1]), theta = c(labels, labels[1]), fill = "toself",
              fillcolor = "rgba(74,111,165,0.25)", line = list(color = "#303983", width = 2.5),
              marker = list(color = "#303983", size = 7)) %>%
        layout(font = pf,
                   polar = list(radialaxis = list(visible = TRUE, range = c(0, 100), gridcolor = "#EBEBEB", linecolor = "#EBEBEB"),
                                angularaxis = list(gridcolor = "#EBEBEB", linecolor = "#EBEBEB", tickfont = list(size = 11, color = "#303983"))),
                   showlegend = FALSE, margin = list(l = 60, r = 60, t = 30, b = 40)) %>% config(displayModeBar = FALSE)
    })

    output$ui_core_metrics <- renderUI({
      df <- get_data(); kpi <- get_kpi()
      kpi_pct <- round(sum(df$PHIBH, na.rm = TRUE) / sum(kpi$DOANH_THU, na.rm = TRUE) * 100, 1)
      renewal <- round(sum(df$LOAI_HD == "Tai tuc") / nrow(df) * 100, 1)
      top_cn <- df %>% filter(TEN_CN != "Chua xac dinh") %>% group_by(TEN_CN) %>% summarise(v = sum(PHIBH)) %>% arrange(desc(v)) %>% slice(1) %>% pull(TEN_CN)
      top_sp <- df %>% filter(TENGOISANPHAM != "Chua xac dinh") %>% group_by(TENGOISANPHAM) %>% summarise(v = sum(PHIBH)) %>% arrange(desc(v)) %>% slice(1) %>% pull(TENGOISANPHAM)
      mr <- function(l, v) tags$tr(
            tags$td(style = "padding:6px 8px; font-weight:bold; border-bottom:1px solid #EBEBEB; width:50%;", l),
            tags$td(style = "padding:6px 8px; border-bottom:1px solid #EBEBEB; color:#303983; font-weight:bold;", v))
      tags$div(style = "font-family:'Helvetica Neue',sans-serif; color:#4A4A4A; font-size:13px;",
        tags$table(style = "width:100%; border-collapse:collapse;",
          mr("Tong GWP", format_vnd(sum(df$PHIBH, na.rm = TRUE))),
          mr("Hoan thanh KPI", paste0(kpi_pct, "%")),
          mr("Phi BH binh quan", format_vnd(mean(df$PHIBH, na.rm = TRUE))),
          mr("Ty le tai tuc", paste0(renewal, "%")),
          mr("CN dan dau", top_cn), mr("SP dan dau", top_sp),
```

```
        mr("Tong goi BH", formatC(nrow(df), format = "d", big.mark = ".")),
        mr("KH unique", formatC(n_distinct(df$MA_KH), format = "d", big.mark = "."))))
  })

  output$ui_conclusion_sales <- renderUI({
    df <- get_data()
    top_cn <- df %>% filter(TEN_CN != "Chua xac dinh") %>% group_by(TEN_CN) %>% summarise(v = sum(PHIBH)) %>% arrange(desc(v)) %>% slice(1)
    tags$div(style = css_b,
      tags$h4(style = css_h, "Hieu suat kinh doanh & Doanh thu"),
      tags$ul(
        tags$li(tags$strong(style = css_s, "Quy mo: "), paste0("Tong ", formatC(nrow(df), format = "d", big.mark = "."), " goi BH voi tong GWP dat ", format_vnd(sum(df$PHIBH, na.rm = TRUE)), ".")),
        tags$li(tags$strong(style = css_s, "Chi nhanh dan dau: "), paste0(top_cn$TEN_CN, " voi ", format_vnd(top_cn$v), "  chiem ty trong lon nhat.")),
        tags$li(tags$strong(style = css_s, "Xu huong: "), "Doanh thu tang truong on dinh tu Q1-2023 den Q2-2025, dau hieu bao hoa nhe o cac thang gan day."),
        tags$li(tags$strong(style = css_s, "Kenh ban: "), "Kenh truc tiep va Dai ly chiem ty trong chi phoi. Kenh E-Commerce va Bancas con nhieu du dia.")))
  })

  output$ui_conclusion_customer <- renderUI({
    df <- get_data()
    renewal <- round(sum(df$LOAI_HD == "Tai tuc") / nrow(df) * 100, 1)
    pct_vip <- round(df %>% filter(XHKH %in% c("Kim cuong", "Bach kim")) %>% pull(MA_KH) %>% n_distinct() / n_distinct(df$MA_KH) * 100, 1)
    tags$div(style = css_b,
      tags$h4(style = css_h, "Hanh vi khach hang"),
      tags$ul(
        tags$li(tags$strong(style = css_s, "Tong KH: "), paste0(formatC(n_distinct(df$MA_KH), format = "d", big.mark = "."), " khach hang  phan bo deu qua 5 hang.")),
        tags$li(tags$strong(style = css_s, "Ty le tai tuc: "), paste0(renewal, "%  can cai thien chien luoc giu chan.")),
        tags$li(tags$strong(style = css_s, "KH gia tri cao: "), paste0(pct_vip, "% thuoc Kim cuong/Bach kim  dong gop phan lon doanh thu.")),
        tags$li(tags$strong(style = css_s, "Cross-sell: "), "So goi TB/KH con thap, dac biet nhom Dong/Bac  co hoi lon.")))
  })

  output$ui_conclusion_product <- renderUI({
    df <- get_data()
    top_sp <- df %>% filter(TENGOISANPHAM != "Chua xac dinh") %>% group_by(TENGOISANPHAM) %>% summarise(v = sum(PHIBH)) %>% arrange(desc(v)) %>% slice(1)
    tags$div(style = css_b,
      tags$h4(style = css_h, "San pham & Danh muc"),
      tags$ul(
        tags$li(tags$strong(style = css_s, "SP dan dau: "), paste0(top_sp$TENGOISANPHAM, " chiem ty trong doanh thu cao nhat.")),
        tags$li(tags$strong(style = css_s, "BH bat buoc: "), "Nhom TNDS xe o to/xe may co volume
```

```
lon nhung phi binh quan thap (Cash Cow)."),
          tags$li(tags$strong(style = css_s, "BH tu nguyen: "), "Nhom Vat chat xe co phi binh quan cao nhat  san pham Star, can day manh."),
          tags$li(tags$strong(style = css_s, "BH con nguoi: "), "Cac goi BHAT, BHSK chiem ty trong nho  tiem nang phat trien qua Bancas.")))
    })

    output$ui_conclusion_challenge <- renderUI({
      tags$div(style = css_b,
        tags$h4(style = css_h, "Thach thuc & Rui ro"),
        tags$ul(
          tags$li(tags$strong(style = css_s, "Tap trung qua muc: "), "Doanh thu phu thuoc lon vao mot so chi nhanh va san pham chu luc  rui ro neu thi truong thay doi."),
          tags$li(tags$strong(style = css_s, "Tai tuc thap: "), "Ty le tai tuc chua toi uu, dac biet o cac chi nhanh tinh  can he thong nhac nho tu dong."),
          tags$li(tags$strong(style = css_s, "Ngoai lai du lieu: "), "Ton tai cac goi BH phi cuc thap (< 1 nghin) va tai khoan Hoi so gop  can kiem tra logic."),
          tags$li(tags$strong(style = css_s, "Kenh moi: "), "E-Commerce va Bancas moi chiem ty trong nho, can dau tu de da dang hoa nguon thu.")))
    })

    output$ui_recom_retention <- renderUI({
      tags$div(style = css_b,
        tags$h4(style = css_h, "Nang cao ty le tai tuc"),
        tags$ul(
          tags$li("Trien khai he thong ", tags$strong("nhac han tu dong"), " qua Zalo/SMS truoc 30-60 ngay het han."),
          tags$li("Chuong trinh ", tags$strong("uu dai tai tuc som"), ": giam 5-10% neu gia han truoc 30 ngay."),
          tags$li("Gan KPI tai tuc rieng cho tung nhan vien va chi nhanh  theo doi hang tuan."),
          tags$li("Chien dich ", tags$strong("win-back"), " cho KH hang Vang+ da het han qua 90 ngay.")))
    })

    output$ui_recom_product <- renderUI({
      tags$div(style = css_b,
        tags$h4(style = css_h, "Chien luoc san pham"),
        tags$ul(
          tags$li("Thiet ke goi ", tags$strong("Combo TNDS + Vat chat"), " cho kenh Dai ly  tang phi binh quan/goi."),
          tags$li("Phat trien goi ", tags$strong("Fleet Shield"), " chuyen biet cho doanh nghiep van tai (cum VIP/K-Means)."),
          tags$li("Mo rong danh muc BH con nguoi (BHAT, BHSK) qua kenh Bancas  phi BQ cao, volume con thap."),
          tags$li("Up-sell KH co xe > 500 trieu nhung phi < 3 trieu  uoc tinh hang nghin leads tiem nang.")))
    })

    output$ui_recom_channel <- renderUI({
      tags$div(style = css_b,
```

```
        tags$h4(style = css_h, "Toi uu kenh phan phoi"),
        tags$ul(
           tags$li("Tang cuong ", tags$strong("kenh E-Commerce"), ": tich hop mua BH online  tiet kiem chi phi phan phoi."),
          tags$li("Day manh ", tags$strong("Bancas"), " voi ngan hang doi tac  KH ngan hang co thu nhap on dinh."),
           tags$li("Chuyen giao mo hinh ", tags$strong("Top Sales"), " tu chi nhanh manh sang chi nhanh yeu."),
           tags$li("Mo rong mang luoi ", tags$strong("gara/salon o to"), "  kenh POS tiep can truc tiep KH mua xe moi.")))
    })

    output$ui_recom_tech <- renderUI({
      tags$div(style = css_b,
        tags$h4(style = css_h, "Cong nghe & Du lieu"),
        tags$ul(
           tags$li("Trien khai ", tags$strong("Goi y san pham"), " vao quy trinh ban hang  goi y san pham realtime."),
            tags$li("Xay dung ", tags$strong("Data Warehouse"), " tap trung  ket noi du lieu ban hang, KH, claim."),
           tags$li("Dashboard ", tags$strong("KPI realtime"), " cho ban lanh dao  theo doi tien do hang ngay."),
           tags$li("Lam sach du lieu ", tags$strong("tai khoan Hoi so"), "  tach rieng giao dich tong cong ty.")))
    })
  })
}
```

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